

METRICS / QUALITY / MODELS / DECISIONS

Data Analytics & Business Intelligence English

Define the measure. Make the finding useful.

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Eight lessons. Eleven extended conversations.

INSIDE THE BOOK

192 vocabulary entries / 128 reusable phrases
220 extended-dialogue turns / 11 transfer exchanges
Word banks and explained answer keys

Upper-intermediate to advanced (B2-C1). For analysts, analytics engineers, reporting specialists, and business partners discussing evidence and decisions.

2026 edition

Eight scenarios. Eight skills.

Eight analytics conversations move from conflicting definitions and data-quality checks to trustworthy calculations, careful inference, and actionable recommendations.

01 Metric Definitions and Business Questions

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Two active-user totals, two different events | Reconcile apparently conflicting metrics by stating the event, distinct entity, window, and denominator.

02 Data Quality and Trust

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Refreshed today, complete only through Monday | Explain mixed source freshness and set an appropriate reporting boundary without mislabeling missing data as zero.

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Training attendees renewed more often | Report a meaningful association while separating the observed difference from an unestablished causal effect.

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The join adds rows, not order value | Explain row grain and join multiplicity, diagnose a duplicated aggregate, and reject a misleading distinct-value fix.

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A positive estimate is not the rollout rule | Present an experiment estimate with its interval, practical threshold, and pre-agreed decision rule.

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The analysis needs weekly counts, not customer names | Specify a purpose-limited data request, distinguish pseudonymization from anonymity, and preserve approval and retention controls.

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Five minutes to authorize the right investigation | Lead a short briefing with a bounded action, quantified evidence, explicit limits, and a review point.

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A number needs a definition and a purpose.

Analytics work often fails in the conversation after the calculation. Teams may use one label for different events, confuse a refresh time with source coverage, or present a promising estimate as a guaranteed effect. These cases supply the facts and practice the language that keeps a useful conclusion attached to its actual evidence.

Agree what is being counted

Name the entity, event, window, and exclusions. A familiar metric label is not a complete definition.

"We counted distinct users with a completed transaction during September."

Put the denominator into speech

A percentage needs an eligible total. Say whether a change is relative or in percentage points.

"That is 480 of 500 cases, or 96%, one point above target."

Keep the limitation near the claim

Qualify source coverage and uncertainty where the result is reported, not only in an appendix.

"The dashboard refreshed today; refunds are complete only through Monday."

End with a bounded recommendation

Connect the finding to an action, owner, and review point without claiming that the cause is already known.

"Approve a two-day investigation of the documented service issue."

PROFESSIONAL BOUNDARIES

Original fictional English practice, not statistical certification, legal advice, a privacy determination, or authorization to use customer data. All datasets, thresholds, local approval rules, and business decisions are invented. Actual analysis must follow the applicable definitions, methods, access controls, and responsible review.

Two active-user totals, two different events

A Read the situation

Reconcile apparently conflicting metrics by stating the event, distinct entity, window, and denominator.

Analyst Mei and product lead Rafael compare September figures for the same 2,000 eligible users, using UTC calendar boundaries and excluding internal and test accounts. One report counts 1,200 distinct users with a successful login. Another counts 900 distinct users with a completed transaction. All 900 are within the login group. Both queries match their stated definitions. The business question is how many eligible users completed a transaction, not how many signed in. The team must choose an accurate label and rate.

PEOPLE / Mei: Data analyst | Rafael: Product lead

TREAT A DISAGREEMENT AS A DEFINITION CHECK

When two teams defend different totals, ask what each number counts before deciding which team made an error. Preserve useful measures with accurate names. A measure can be correctly calculated yet poorly suited to the business question on the table.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which count answers the supplied business question?

- | | |
|--|--|
| A 900 distinct users with a completed transaction | B 1,200 distinct users with a login |
| C 2,100 users obtained by adding the groups | D 2,000 completed transactions |

2. What is the correct eligible-user transaction rate?

- | | |
|--------------|---------------|
| A 75% | B 105% |
| C 45% | D 60% |

3. Why can both query results be correct?

- | | |
|--|--|
| A Every metric must have the same count. | B The larger number is always more accurate. |
| C A login and a transaction are identical events. | D They apply different event definitions to the same eligible population. |

[Answers and explanations: page 85.](#)

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

metric

A defined quantitative measure.

IN USE define a metric

business question

The decision-relevant question an analysis is meant to answer.

IN USE clarify the business question

metric definition

Rules specifying exactly how a measure is calculated.

IN USE agree a metric definition

qualifying event

An action meeting the condition for inclusion.

IN USE specify the qualifying event

distinct user

A unique user counted once under a defined identity rule.

IN USE count distinct users

event count

The number of recorded qualifying occurrences.

IN USE distinguish event counts from users

eligible population

The complete group allowed into the measure's base.

IN USE define the eligible population

denominator

The total used as the base of a rate.

IN USE state the denominator

numerator

The counted outcome placed over the base of a rate.

IN USE verify the numerator

reporting window

The time period included in a measure.

IN USE align reporting windows

calendar month

A month bounded by the adopted calendar and time zone.

IN USE report by calendar month

time zone

The time reference used to assign event dates.

IN USE fix the reporting time zone

Precision check

900/2,000 is 45% of eligible users. 900/1,200 is 75% of users who logged in. Both calculations are valid for their stated bases, but only the first answers the supplied eligible-user question.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

exclusion rule

A condition removing records from the measure.

IN USE document exclusion rules

internal account

An account belonging to the organization's own personnel or systems.

IN USE exclude internal accounts

test account

An account used for testing rather than genuine activity.

IN USE filter test accounts

deduplication

Removing repeated representations of the same counted entity.

IN USE apply user deduplication

identity resolution

Connecting records that refer to the same entity.

IN USE review identity resolution

active user

A user meeting a specified activity definition.

IN USE define active users explicitly

login-based activity

Activity defined by a qualifying sign-in.

IN USE report login-based activity

transaction-based activity

Activity defined by a qualifying transaction.

IN USE report transaction-based activity

subset

A group entirely contained within another group.

IN USE identify the subset relationship

metric owner

The role accountable for a measure's definition and use.

IN USE identify the metric owner

semantic layer

Shared definitions connecting data to business meaning.

IN USE maintain a semantic layer

reconciliation

Explaining and resolving differences between reported figures.

IN USE reconcile metric totals

Precision check

The two groups overlap: all 900 transacting users are among the 1,200 login users. Adding them would double-count those 900. Repeated transactions by one person also do not increase a distinct-user total.

Phrases you can actually use

C Notice the language / 1 of 2

CLARIFY THE QUESTION

"Are we measuring sign-in or completed transactions?"

NAME THE EVENT

"The qualifying event is a completed transaction."

KEEP EXCLUSIONS ALIGNED

"Both measures exclude internal and test accounts."

EXPLAIN THE DIFFERENCE

"The reports use different activity events."

DEFINE THE COUNTED ENTITY

"Each user is counted once."

SET THE WINDOW

"Use the September calendar month in UTC."

ACKNOWLEDGE BOTH RESULTS

"Both queries match their own definitions."

CHOOSE THE RELEVANT COUNT

"Nine hundred users answer the transaction question."

Active

Incomplete unless the qualifying event is stated.

Users

Distinct identities, not the number of actions.

Of eligible users

Fixes the denominator for the rate.

Phrases you can actually use

C Notice the language / 2 of 2

STATE THE BASE

"The eligible population is 2,000 users."

DISTINGUISH ANOTHER RATE

"Seventy-five percent uses logged-in users as its base."

IMPROVE THE LABEL

"Call this transaction-active users."

ASSIGN DEFINITION OWNERSHIP

"The metric owner should approve the shared definition."

GIVE THE RATE

"That is 45% of eligible users."

AVOID DOUBLE-COUNTING

"The transaction group is a subset of the login group."

PRESERVE THE OTHER MEASURE

"Keep the login measure under its own label."

CLOSE THE DISAGREEMENT

"We need consistent meaning, not an artificial match between different events."

During September

Needs consistent calendar boundaries and time zone.

Correct

Can mean calculated correctly without being decision-relevant.

Reconcile

Explain differences before forcing totals to match.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which label best fits the 900 count?

- A** September transaction-active users
- B** All September logins
- C** Completed transaction events, including repeats
- D** Total eligible accounts

2. What does 900/1,200 measure?

- A** A 75% growth rate from last month
- B** The share of logged-in users who completed a transaction
- C** The share of all eligible users who logged in
- D** The number of repeated transactions

3. Why should the two totals not be added?

- A** Each report contains exactly the same 1,200 people.
- B** The 900 transacting users are already included among the 1,200 login users.
- C** All additions of user counts are invalid.
- D** The UTC boundary changes addition rules.

4. Which next step addresses the real disagreement?

- A** Alter a correct query until both totals match.
- B** Use whichever number looks more impressive.
- C** Remove the denominator from the slide.
- D** Record both event definitions and select the transaction measure for this question.

Answers and explanations: [page 85](#).

Two active-user totals, two different events

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

subset / semantic layer / distinct user / numerator / metric owner / qualifying event / reconciliation / denominator / business question / reporting window

- 01 **Rafael:** The executive slide says twelve hundred active users, but Finance has nine hundred. Both teams insist their query is right. Which number should I remove?
- 02 **Mei:** First agree the ¹ _____. We need the number of eligible users who completed a transaction during September, not merely the number who signed in.
- 03 **Rafael:** The twelve hundred report counts successful logins, once per user. Finance counts completed transactions, also once per user. Both exclude internal and test accounts.
- 04 **Mei:** Then the ² _____ differs. The queries can both be correct while answering different questions. We should not change a valid calculation simply to make two different meanings agree.
- 05 **Rafael:** The window is the September calendar month in UTC for both, and they use the same two thousand eligible users. That part is aligned.
- 06 **Mei:** Good. A shared ³ _____ removes one possible source of disagreement. The distinction here is activity type, not different month boundaries or different treatment of test accounts.
- 07 **Rafael:** For the actual question, I should choose nine hundred and label it as people completing a transaction. Multiple transactions by one user still count once.
- 08 **Mei:** Yes, the entity is a ⁴ _____. That is not the total number of transaction events. The slide should make the counted entity clear enough that the audience cannot silently substitute another.
- 09 **Rafael:** I have seventy-five percent on the slide: nine hundred divided by twelve hundred. Is that answering the eligible-user question, or have I changed the base?
- 10 **Mei:** That uses a different ⁵ _____. Nine hundred divided by two thousand eligible users is forty-five percent. Nine hundred divided by twelve hundred logged-in users is seventy-five percent.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 10.

- 11 **Rafael:** Both rates tell us something, but forty-five percent is the answer to the eligible-user question. Seventy-five percent describes the logged-in group instead.
- 12 **Mei:** Exactly. Keep the ⁶ _____ and base together in speech. Saying just the percentage would hide the change in meaning and invite an incorrect comparison later.
- 13 **Rafael:** Someone suggested adding the two reports to show total engagement. But the nine hundred transacting users are already included in the twelve hundred who logged in.
- 14 **Mei:** That ⁷ _____ relationship makes addition inappropriate for distinct people. It would count the transaction group twice rather than identify twenty-one hundred different active users.
- 15 **Rafael:** I will keep the login figure because it is useful, but give it a different label. The executive headline will use the transaction definition.
- 16 **Mei:** Put both in the ⁸ _____ with the event, identity rule, exclusions, and time zone. Renaming one slide is not enough if other reports keep reusing the ambiguous active-user label.
- 17 **Rafael:** Who signs off if the transaction event changes next month? I do not want the same label to hide a new definition.
- 18 **Mei:** Assign a ⁹ _____ and record the approved version. That gives both teams a place to resolve changes before comparing totals that no longer mean the same thing.
- 19 **Rafael:** My summary will be nine hundred transaction-active users, forty-five percent of two thousand eligible users, September UTC. The login measure remains separately labeled.
- 20 **Mei:** That completes the ¹⁰ _____. We have preserved both correct calculations, selected the measure that answers the decision, and removed the ambiguity that created the apparent conflict.

ANSWERS / EXPLANATIONS: PAGE 86

1. business question | 2. qualifying event | 3. reporting window | 4. distinct user | 5. denominator | 6. numerator
| 7. subset | 8. semantic layer | 9. metric owner | 10. reconciliation

One count, two possible bases

F Say it

A month has 1,000 eligible users, 800 distinct login users, and 400 distinct transacting users, all within the login group. The question asks for transaction activity among all eligible users.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

1000 / 40 / 50 / 400

Analyst: "The relevant distinct-user count is ¹ _____."

Manager: "The eligible-user denominator is ² _____."

Analyst: "The requested rate is ³ _____ percent."

Manager: "Using login users instead would give ⁴ _____ percent."

ANSWERS / EXPLANATIONS: PAGE 86

1. 400 | 2. 1000 | 3. 40 | 4. 50

Rehearse with a fixed script

1. Read turns 1-10. Stress qualifying event and denominator as separate choices.
2. Switch roles for turns 11-20. Keep the subset relationship clear when rejecting addition.
3. Check the transfer. Read 40% and 50% with their different population bases.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Refreshed today, complete only through Monday

A Read the situation

Explain mixed source freshness and set an appropriate reporting boundary without mislabeling missing data as zero.

Reporting analyst Asha reviews a dashboard with operations manager Tomas on Wednesday at 09:00 UTC. The dashboard refresh completed, and the orders source is verified complete through Tuesday 23:59 UTC. The refunds source is verified complete only through Monday 23:59 UTC because its next delivery is delayed. A Monday-Tuesday net-order-value figure subtracts available refunds from orders, so the periods are misaligned. Asha can publish a verified Monday-only comparison; Tuesday refunds are unknown, not confirmed zero. No delivery time for the missing source is confirmed.

PEOPLE / Asha: Reporting analyst | Tomas: Operations manager

EXPLAIN THE AFFECTED DECISION, NOT JUST THE PIPELINE

A completed refresh can sound like an all-clear to a business reader. Name the source that is late, the measure affected, and the period that remains usable. This gives colleagues a practical reporting option while preserving the visible limitation.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which source is complete only through Monday?

A Refunds

B Orders

C Both sources through Tuesday

D Neither source through any date

2. What is known about Tuesday refunds?

A They are confirmed zero.

B They equal Tuesday orders.

C They were all processed on Monday.

D They are unknown in the available source.

3. Which comparison can Asha publish as verified?

A A Wednesday full-day total

B A guaranteed delivery time for the missing source

C Monday-only, using aligned source coverage

D Monday-Tuesday net value without qualification

Answers and explanations: page 87.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

data quality

Fitness of data for its intended use.

IN USE assess data quality

freshness

How recent data is relative to the intended use.

IN USE monitor data freshness

timeliness

Availability of data within the required time.

IN USE define timeliness requirements

completeness

Presence of the required records or values.

IN USE verify source completeness

accuracy

Agreement between data and the facts it represents.

IN USE assess data accuracy

validity

Conformance to defined formats or rules.

IN USE test data validity

consistency

Agreement across related representations or rules.

IN USE check data consistency

uniqueness

Absence of unintended duplicate records.

IN USE test record uniqueness

refresh timestamp

The time a report or model was last updated.

IN USE display the refresh timestamp

source cutoff

The boundary through which source data is available or complete.

IN USE state the source cutoff

event time

When the represented business event occurred.

IN USE distinguish event time from load time

load time

When data entered the relevant system.

IN USE record load time

Precision check

A refresh timestamp records when processing ran, not how current every input was. Even a latest event timestamp alone would not prove completeness. Here, the brief explicitly verifies different complete-through cutoffs for the two sources.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

upstream source

An earlier system supplying data to a pipeline.

IN USE inspect upstream sources

pipeline

A sequence of data movement and transformation steps.

IN USE monitor the data pipeline

ingestion delay

A lag in receiving data from a source.

IN USE investigate ingestion delay

late-arriving data

Records received after the expected processing point.

IN USE handle late-arriving data

backfill

Loading previously missing historical records.

IN USE validate a backfill

watermark

A tracked boundary used to process or assess data progress.

IN USE inspect the pipeline watermark

coverage window

The period or population represented by available data.

IN USE align coverage windows

reconciliation check

A comparison used to verify agreement between records or totals.

IN USE run reconciliation checks

null value

A marker for missing or unknown information in a data field.

IN USE distinguish null from zero

zero value

A recorded quantity of none, not automatically a missing value.

IN USE preserve zero values correctly

data-quality notice

A visible explanation of a known data limitation.

IN USE publish a data-quality notice

recovery estimate

A qualified expectation for restoration of a data flow.

IN USE qualify the recovery estimate

Precision check

Unknown Tuesday refunds must not be presented as zero refunds. A combined Monday-Tuesday net figure would mix different coverage windows. A verified Monday-only view is a bounded alternative while the missing delivery is unresolved.

Phrases you can actually use

C Notice the language / 1 of 2

ACKNOWLEDGE THE REFRESH

"The dashboard refresh completed at 09:00 UTC."

NAME THE LAGGING SOURCE

"Refunds are complete only through Monday."

LOCATE THE AFFECTED MEASURE

"The two-day net figure mixes coverage periods."

OFFER A USABLE VIEW

"We can publish the verified Monday-only comparison."

SEPARATE SOURCE COVERAGE

"That does not mean every source is current through Tuesday."

STATE THE OTHER CUTOFF

"Orders are verified through Tuesday 23:59 UTC."

DISTINGUISH MISSING FROM ZERO

"Tuesday refunds are unknown, not confirmed zero."

KEEP THE LIMITATION VISIBLE

"Place the source-cutoff note beside the figure."

Fresh

Relative to a defined required time, not merely a green icon.

Refreshed

Describes processing, not necessarily complete source coverage.

Through Monday

State the actual cutoff and time zone.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID A FALSE PROMISE

"The missing delivery time is not confirmed."

DESCRIBE THE DEPENDENCY

"The calculation depends on both orders and refunds."

PRESERVE THE OLD STATUS

"Keep the affected view marked incomplete until validation passes."

BOUND THE CURRENT CLAIM

"This view is complete through Monday only."

ASK FOR THE RIGHT EVIDENCE

"Check source completeness, not only job completion."

NAME THE NEXT VALIDATION

"Reconcile the backfill before republishing the two-day figure."

CLARIFY TIMING

"Load time and event time are different."

CLOSE THE UPDATE

"Publish the aligned period and report the remaining source delay."

Unknown

Must not silently become zero.

Backfilled

Requires validation before the limitation is removed.

Expected

An estimate needing a basis, not a delivery guarantee.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which dashboard note is accurate?

- A** Refreshed Wednesday 09:00 UTC; refunds complete through Monday; two-day net value incomplete.
- B** Refreshed Wednesday, so all source events are current.
- C** Tuesday refunds are zero because none arrived.
- D** The refund delay proves every order is inaccurate.

2. Why is Monday-only reporting usable here?

- A** Monday is always more accurate than Tuesday.
- B** Removing refunds would make any period complete.
- C** A short period never needs validation.
- D** Both sources are verified complete through that shared period.

3. Which check is insufficient by itself?

- A** Seeing that the dashboard refresh job completed
- B** Verifying source coverage against expected records
- C** Reconciling the missing source after backfill
- D** Checking the complete-through cutoff for each input

4. What should happen after the delayed records arrive?

- A** Validate and reconcile them before removing the limitation.
- B** Immediately claim that every historical figure was already correct.
- C** Keep treating the records as zero.
- D** Change the event dates to Wednesday load time.

Answers and explanations: [page 87](#).

Refreshed today, complete only through Monday

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

coverage window / completeness / timeliness / recovery estimate / source cutoff / event time / refresh timestamp / data-quality notice / zero value / backfill

- 01 **Tomas:** The dashboard says refreshed this morning. Can I use the Monday-Tuesday net-order-value number in the operations meeting as our complete two-day result?
- 02 **Asha:** Not yet. The ¹ _____ is Wednesday at nine UTC, but the inputs have different coverage. Orders are verified through Tuesday; refunds are complete only through Monday.
- 03 **Tomas:** I had assumed that a successful refresh meant every table contained the latest business activity. Which boundary should I read instead for the refund data?
- 04 **Asha:** Read the ² _____. Refunds are complete through Monday at twenty-three fifty-nine UTC. The next delivery is delayed, so Tuesday is not covered by the available records.
- 05 **Tomas:** Then the two-day net figure has Tuesday orders but not the corresponding Tuesday refunds. It could look better simply because that input is incomplete.
- 06 **Asha:** Yes. The ³ _____ is misaligned. We cannot interpret the result as a complete two-day net amount when the subtraction uses refunds from a shorter period.
- 07 **Tomas:** The chart fills the empty Tuesday cell with zero. Should we change that to unknown while the refund delivery is missing?
- 08 **Asha:** Missing is not a ⁴ _____. We do not know the amount yet. Showing zero as the business result would turn a source limitation into a false claim about customer activity.
- 09 **Tomas:** I still need a reliable number for the meeting. Could we show Monday by itself, since both sources are verified complete through that day?
- 10 **Asha:** Yes. That respects ⁵ _____ for the shared period in this case. We can publish the Monday-only comparison and make clear that the two-day figure remains incomplete.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 19.

- 11 **Tomas:** The note should be near the result, not somewhere people will miss it. I will explain that Tuesday orders alone do not settle the net figure.
- 12 **Asha:** Add a ⁶ _____ beside the affected view. State which source is delayed, its complete-through time, and which period remains usable rather than labeling the entire dashboard unreliable.
- 13 **Tomas:** Do we know when the missing refund delivery will arrive? I would like to tell the meeting whether the full figure will be available this afternoon.
- 14 **Asha:** There is no confirmed ⁷ _____. I will request an update from the source owner, but I should not invent a delivery time simply because the meeting needs one.
- 15 **Tomas:** Once the rows arrive, we will have business events from Tuesday loaded on a later day. We should not move those refunds into Wednesday totals.
- 16 **Asha:** Correct. Preserve ⁸ _____ separately from load time. The records belong to the appropriate business period even when the pipeline receives them later than expected.
- 17 **Tomas:** When the delivery arrives, who checks that it is complete? I do not want the warning removed just because the refresh turns green.
- 18 **Asha:** Validate the ⁹ _____ and document the checks. A green refresh icon should not be our only evidence that the previously incomplete reporting period is now fit for use.
- 19 **Tomas:** I will take the Monday-only view to the meeting, explain the Tuesday refund gap, and avoid promising a two-day net result before it is verified.
- 20 **Asha:** That is the right ¹⁰ _____ message: a useful aligned result now, a specific limitation, and a clear validation step before the fuller figure is released.

ANSWERS / EXPLANATIONS: PAGE 88

1. refresh timestamp | 2. source cutoff | 3. coverage window | 4. zero value | 5. completeness | 6. data-quality notice | 7. recovery estimate | 8. event time | 9. backfill | 10. timeliness

A current job can use an old source

F Say it

A report refreshed Friday. Sales are verified complete through Thursday, but returns only through Wednesday. Thursday returns are unknown. Both sources are complete for Wednesday.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

Friday / unknown / Wednesday / aligned

Analyst: "The refresh ran on ¹ _____."

Manager: "Returns are complete only through ² _____."

Analyst: "Thursday returns are ³ _____."

Manager: "A Wednesday-only comparison is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 88

1. Friday | 2. Wednesday | 3. unknown | 4. aligned

Rehearse with a fixed script

1. Read turns 1-10. Contrast the refresh timestamp with each source cutoff.
2. Switch roles for turns 11-20. Stress event time rather than the later load time.
3. Check the transfer. Read the Wednesday-only option without turning Thursday unknowns into zeros.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Four hundred eighty needs a denominator and a target

A Read the situation

Turn an isolated monthly count into a clear performance statement with definition, target, comparison, and units.

BI analyst Jonas reviews a card labeled 480 with service director Elena. It counts September requests completed within 24 hours of receipt: 480 of 500 eligible requests. All requests have a complete observation window, and the remaining 20 finished late. The approved target is at least 95%. August had 450 timely completions among 500 eligible requests under the same definition. Jonas must report the current rate, position against target, and change from August without calling a percentage-point change a relative percentage increase.

PEOPLE / Jonas: Business-intelligence analyst | Elena: Service director

MAKE THE INTERPRETATION PORTABLE

A useful executive card should survive being quoted outside the meeting. Include the measure and comparison in a sentence that can be repeated accurately. Avoid forcing the listener to infer whether a larger number represents improvement, demand growth, or a changed definition.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the September timely-completion rate?

- A 96%
- C 90%

- B 95%
- D 480%

2. How does September compare with the target?

- A Exactly equal to target
- C One percent below target

- B One percentage point above the 95% target
- D Six percentage points below target

3. What was the August rate?

- A 45%
- C 96%

- B 95%
- D 90%

Answers and explanations: [page 89](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

KPI

Key performance indicator, a measure tied to an objective.

IN USE define a KPI

dashboard card

A compact display of a selected metric.

IN USE label the dashboard card

performance target

A specified desired level of a measure.

IN USE compare with the performance target

actual result

The observed value of the defined measure.

IN USE report the actual result

comparison period

An earlier or alternative interval used as a reference.

IN USE state the comparison period

variance

A difference from a specified reference value.

IN USE explain the target variance

percentage point

One unit of difference between percentage values.

IN USE report percentage-point changes

relative change

Change divided by the original reference value.

IN USE calculate relative change

month-on-month

Compared with the immediately preceding month.

IN USE describe month-on-month movement

year-on-year

Compared with the corresponding period a year earlier.

IN USE report year-on-year change

service-level target

An agreed performance threshold for a service measure.

IN USE track the service-level target

timely completion

Completion within the specified permitted duration.

IN USE measure timely completion

Precision check

September is 96%: 480/500. August is 90%: 450/500. The increase is 6 percentage points, or approximately 6.67% relative to August. September exceeds the 95% target by 1 percentage point.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

elapsed time

Duration between defined start and end events.

IN USE calculate elapsed time

observation window

The period available to observe the relevant outcome.

IN USE complete the observation window

threshold

A boundary used for a rule or decision.

IN USE apply the threshold

direction of improvement

Whether higher or lower values are favorable.

IN USE state the direction of improvement

trend

A pattern across a sequence of observations.

IN USE qualify a trend claim

seasonality

Recurring variation related to calendar patterns.

IN USE assess seasonality

run chart

A display of values ordered through time.

IN USE review the run chart

annotation

Explanatory text attached to a chart or figure.

IN USE add a relevant annotation

drill-down

Moving from a summary to more detailed data.

IN USE provide a drill-down

exception queue

Items selected for attention under a defined rule.

IN USE review the exception queue

comparability

The extent to which measures use compatible bases.

IN USE establish period comparability

executive summary

A concise account of the main evidence and decision relevance.

IN USE sharpen the executive summary

Precision check

The 480 card alone omits its denominator, timing rule, target, and comparison. Two monthly results establish a month-on-month improvement here, not a sustained long-term trend or an explanation of why performance changed.

Phrases you can actually use

C Notice the language / 1 of 2

03

NAME THE MEASURE

"This is the share completed within 24 hours."

GIVE COUNT AND BASE

"September recorded 480 timely completions out of 500."

COMPARE WITH TARGET

"That is one percentage point above the 95% target."

USE THE CORRECT CHANGE UNIT

"The rate rose by six percentage points."

STATE THE START

"The clock starts when the request is received."

STATE THE RESULT

"The timely-completion rate is 96%."

GIVE THE PRIOR RESULT

"August was 450 of 500, or 90%."

GIVE THE RELATIVE ALTERNATIVE

"That is about a 6.67% relative increase."

Above target

State the amount and use the correct unit.

Up six points

Not identical to up six percent.

Within 24 hours

Requires defined start and completion events.

Phrases you can actually use

C Notice the language / 2 of 2

03

PROTECT COMPARABILITY

"Both months use the same definition."

KEEP THE EXCEPTIONS VISIBLE

"Twenty September requests finished late."

BOUND THE TREND

"This is a two-month comparison, not a sustained-trend claim."

OFFER THE NEXT VIEW

"The drill-down can show the late requests."

CONFIRM MATURITY

"Every request has a complete observation window."

AVOID AN INVENTED CAUSE

"These totals do not explain the improvement."

MAKE THE CARD INTERPRETABLE

"Add the definition, denominator, target, and comparison."

CLOSE THE READOUT

"September met the target and improved from August on the same basis."

Monthly total

Does not by itself reveal a performance rate.

Trend

Needs more than a confident adjective about two observations.

Favorable

Depends on the measure's direction and target.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which card text is most complete?

- A** 96, target unknown.
- B** Up 6%, with no reference period.
- C** September: 480/500 within 24 hours, 96%; target 95%; August 90%.
- D** 480, excellent.

2. Which change statement is accurate?

- A** Up 6% relative to August exactly
- B** Up 30 percentage points because the count rose by 30
- C** Up 1 percentage point from August
- D** Up 6 percentage points from August

3. What is the approximate relative increase from 90% to 96%?

- A** 6 percentage points expressed as 60%
- B** 1%
- C** 96%
- D** 6.67%

4. What cannot be concluded from the supplied totals?

- A** The two months used the same definition
- B** The specific cause of the improvement
- C** September met the target
- D** Twenty September requests finished late

Answers and explanations: [page 89](#).

Four hundred eighty needs a denominator and a target

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

KPI / actual result / comparability / percentage point / observation window / executive summary / relative change / drill-down / comparison period / month-on-month

- 01 **Elena:** The card says four hundred eighty with a green arrow. What does it count, and how should I interpret it?
- 02 **Jonas:** The ¹ _____ is timely completion: requests finished within twenty-four hours of receipt. Four hundred eighty is the timely count, not the rate or the total demand.
- 03 **Elena:** There were five hundred eligible requests in September, and the other twenty finished late. All requests have had enough time for us to observe their outcomes.
- 04 **Jonas:** Then the ² _____ is ninety-six percent, or four hundred eighty out of five hundred. We should display both the count and the base so the rate is interpretable.
- 05 **Elena:** Our approved target is at least ninety-five percent. That means September passed, but I want the amount above the target described accurately.
- 06 **Jonas:** It is one ³ _____ above target. Saying one percent without qualification can suggest a relative calculation, which is a different way of expressing the comparison.
- 07 **Elena:** August had four hundred fifty timely requests out of five hundred. The same definition and exclusions apply, so that earlier rate was ninety percent.
- 08 **Jonas:** August is the ⁴ _____. September increased by six percentage points on that basis. We should name the month rather than leave the audience guessing what the arrow compares.
- 09 **Elena:** My draft says up six percent. Should I change that to six percentage points? I need wording I can repeat without changing the calculation.
- 10 **Jonas:** Correct it. The ⁵ _____ is approximately six point six seven percent. The absolute difference between the percentage values is six points. Both are valid when labeled, but they are not interchangeable.

The conversation, continued

E Conversations / turns 11-20

03

Continue with the word bank on page 28.

- 11 **Elena:** I also want to avoid the green arrow hiding the twenty late requests. Meeting the overall threshold does not mean every customer received the target service.
- 12 **Jonas:** We can provide a ⁶ _____ for those late cases. The headline can report that the target was met while the detail preserves the exceptions that still need operational attention.
- 13 **Elena:** Can I call it a sustained improvement, or do two months only support the month-on-month comparison? I want to recognize the result accurately.
- 14 **Jonas:** We can report the ⁷ _____ improvement. Two months do not establish a sustained trend, and these totals do not identify the cause. Recognition does not require a stronger statistical story.
- 15 **Elena:** Fair. We can acknowledge the result without attributing it to a particular staffing change that has not been evaluated. The definition should stay visible too.
- 16 **Jonas:** Preserve ⁸ _____ by retaining the same start, completion rule, exclusions, and denominator. A future definition change should be flagged instead of being presented as an unexplained performance movement.
- 17 **Elena:** The clock begins on receipt and stops on completion. We should also state that the data covers fully observed requests, not a partial month of immature outcomes.
- 18 **Jonas:** Yes, the ⁹ _____ is complete for every request here. That matters because unresolved timing can otherwise distort a rate even when the arithmetic itself is correct.
- 19 **Elena:** September was ninety-six percent: one point above target, six above August, twenty requests late. The cause is unestablished.
- 20 **Jonas:** That is a useful ¹⁰ _____. The card will include the definition and base, and the detail will support the late-case review rather than asking the audience to infer everything from 480.

ANSWERS / EXPLANATIONS: PAGE 90

1. KPI | 2. actual result | 3. percentage point | 4. comparison period | 5. relative change | 6. drill-down | 7. month-on-month | 8. comparability | 9. observation window | 10. executive summary

Above target and above last month are different

F Say it

A service completes 180 of 200 eligible requests on time this month. Last month was 170 of 200. The target is 88%, and both months use the same fully observed definition.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

five / 90 / 85 / two

Analyst: "This month is ¹ _____ percent."

Manager: "Last month was ² _____ percent."

Analyst: "The month-on-month increase is ³ _____ percentage points."

Manager: "This month exceeds target by ⁴ _____ percentage points."

ANSWERS / EXPLANATIONS: PAGE 90

1. 90 | 2. 85 | 3. five | 4. two

Rehearse with a fixed script

1. Read turns 1-10. Say percentage points and relative percent distinctly.
2. Switch roles for turns 11-20. Keep the twenty late requests in the summary.
3. Check the transfer. Contrast five points above last month with two above target.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Training attendees renewed more often

A Read the situation

04

Report a meaningful association while separating the observed difference from an unestablished causal effect.

Analyst Priya reviews voluntary customer-training results with customer-success lead Marcus. Over the same renewal window, 90 of 100 eligible attendees renewed, compared with 160 of 200 eligible non-attendees. Attendance was not randomized. Baseline motivation and engagement may differ between the groups and were not measured. The records establish a 90% versus 80% renewal association, not the effect of making everyone attend. Priya recommends a reviewed evaluation design before the team uses the difference as a causal benefit claim.

PEOPLE / Priya: Data analyst | Marcus: Customer-success lead

PROTECT THE USEFUL OBSERVATION WHILE CORRECTING THE CLAIM

A causal caveat should not sound like all the work was pointless. State the real association first, then explain the specific competing explanation. This helps a colleague keep the finding while replacing an overconfident promise with an appropriate next evaluation.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the attendee renewal rate?

A 80%

B 45%

C 10%

D 90%

2. What is the non-attendee renewal rate?

A 90%

B 20%

C 80%

D 160%

3. Why is the causal effect not established?

A The renewal window differs by definition.

B Attendance was voluntary, and unmeasured baseline differences may matter.

C The two rates are identical.

D No customer renewed.

Answers and explanations: page 91.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

association

An observed relationship between variables or groups.

IN USE report an association

correlation

A measure or description of co-variation.

IN USE distinguish correlation from causation

causal effect

A change attributable to a specified intervention.

IN USE estimate a causal effect

confounding

Distortion from factors related to both exposure and outcome.

IN USE assess potential confounding

self-selection

Participation determined partly by people's own choices.

IN USE account for self-selection

selection bias

Systematic distortion related to how observations enter groups.

IN USE evaluate selection bias

baseline characteristic

A feature measured before the exposure or intervention.

IN USE compare baseline characteristics

motivation

A person's underlying inclination to act or engage.

IN USE assess baseline motivation

engagement

Participation or involvement under a specified definition.

IN USE define customer engagement

observational study

Analysis of naturally occurring exposures without assigned treatment.

IN USE interpret an observational study

random assignment

Allocation to groups by a defined random process.

IN USE use random assignment

treatment group

The group assigned or exposed to the studied intervention.

IN USE define the treatment group

Precision check

The observed difference is 10 percentage points: 90%-80%. Relative to the non-attendee rate, it is 12.5%. Neither expression turns the voluntary comparison into an estimated causal benefit for mandatory training.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

comparison group

The group used as a reference for the outcome.

IN USE select a comparison group

counterfactual

The outcome that would occur under an alternative exposure.

IN USE define the counterfactual question

renewal

Continuation of an eligible customer contract or relationship.

IN USE measure customer renewal

renewal window

The period over which renewal outcomes are observed.

IN USE align renewal windows

eligible customer

A customer meeting the defined inclusion conditions.

IN USE identify eligible customers

absolute difference

A difference expressed in the original measure's units.

IN USE state the absolute difference

relative lift

A difference expressed relative to a comparison value.

IN USE qualify relative lift

adjustment

Accounting for specified variables in an analysis.

IN USE assess the adjustment method

unmeasured factor

A relevant characteristic not recorded in the data.

IN USE disclose unmeasured factors

generalizability

Applicability of a result beyond the studied group.

IN USE assess generalizability

causal claim

A statement that an intervention produced an outcome.

IN USE qualify a causal claim

evaluation design

The planned method for assessing a program or intervention.

IN USE review the evaluation design

Precision check

Unmeasured motivation is a possible confounder, not a proven explanation of the full difference. Likewise, failure to establish causation does not prove that training has no effect. The appropriate conclusion is narrower than either claim.

Phrases you can actually use

C Notice the language / 1 of 2

STATE THE OBSERVED RATES

"Attendees renewed at 90%; non-attendees at 80%."

GIVE THE COUNTS

"That is 90 of 100 versus 160 of 200."

NAME THE DIFFERENCE

"The observed gap is ten percentage points."

LABEL THE FINDING

"This is an association in the supplied data."

IDENTIFY SELECTION

"Customers chose whether to attend."

NAME THE ALTERNATIVE EXPLANATION

"Baseline engagement may differ between the groups."

PRESERVE UNCERTAINTY

"Those baseline factors were not measured."

REJECT THE CAUSAL LEAP

"We cannot attribute the whole difference to training."

Because

Introduces a causal claim that needs appropriate support.

Among

Describes a group comparison without asserting cause.

May differ

Identifies a plausible concern, not a proven explanation.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID THE OPPOSITE OVERCLAIM

"This does not prove that training has no effect."

KEEP THE WINDOW ALIGNED

"Both rates use the same renewal window."

QUALIFY ADJUSTMENT

"Adjusting measured factors would not automatically remove every unmeasured difference."

REVISE THE HEADLINE

"Use higher observed renewal among attendees."

CLARIFY THE COUNTERFACTUAL

"We need to know what would have happened to comparable customers without training."

BOUND THE AUDIENCE

"The result concerns these eligible customer groups."

RECOMMEND EVALUATION

"Review a design that can better isolate the training effect."

CLOSE THE DISCUSSION

"Keep the association and remove the unsupported promise."

Lift

Specify whether descriptive or causal, relative or absolute.

No proof of effect

Not equivalent to proof of no effect.

Everyone

Extends the claim beyond the observed voluntary groups.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which headline is supported?

- A** Mandatory training is proven to prevent all churn
- B** Higher observed renewal among voluntary training attendees
- C** Training caused a ten-point renewal increase for everyone
- D** Training has no possible effect

2. What is the relative observed difference compared with 80%?

- A** 12.5%
- B** 10% exactly
- C** 90%
- D** 20 percentage points

3. Which statement about motivation is accurate?

- A** It is proven to explain the entire gap.
- B** It was balanced by random assignment.
- C** It was measured and found identical.
- D** It is an unmeasured possible source of group differences.

4. What should a revised recommendation do?

- A** Preserve the association and seek a reviewed design for a causal evaluation.
- B** Hide the group sizes and announce a guaranteed benefit.
- C** Treat voluntary attendance as random assignment.
- D** Conclude no evaluation can ever be useful.

Answers and explanations: [page 91](#).

Training attendees renewed more often

E Conversations / turns 1-10

04

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

unmeasured factor / evaluation design / association / causal effect / relative lift / counterfactual / adjustment / Confounding / generalizability / self-selection

- 01 **Marcus:** Training attendees renewed at ninety percent, versus eighty percent for non-attendees. Can we say the sessions improved renewal by ten points?
- 02 **Priya:** We can report the ¹ _____, but not that causal conclusion from these records. Ninety of one hundred attendees renewed; one hundred sixty of two hundred non-attendees renewed.
- 03 **Marcus:** Both groups had the same renewal window and eligibility rules. That makes the comparison cleaner, even though attendance was voluntary rather than assigned.
- 04 **Priya:** Correct, but ² _____ remains important. Customers who chose training may have been more motivated or engaged already. Those baseline differences were not measured in the supplied data.
- 05 **Marcus:** Then the ten-point difference could reflect training, pre-existing differences, or both. We should not assume the entire gap was created by the sessions.
- 06 **Priya:** Exactly. The ³ _____ asks what the intervention changes compared with what would otherwise have happened. This comparison does not isolate that quantity simply because the rates are clearly different.
- 07 **Marcus:** I do not want the caveat to erase the positive finding. The team has real evidence that attendees renewed more often in this group.
- 08 **Priya:** Keep that observation. ⁴ _____ is a reason to qualify the causal interpretation, not to deny the counts. We can state the association and the specific limitation together.
- 09 **Marcus:** Sales will ask whether you are saying motivation caused the whole gap. We did not measure it, so that is not our conclusion either, is it?
- 10 **Priya:** Right. It is an ⁵ _____ that may matter, not a proven explanation of every percentage point. We should avoid replacing one unsupported causal story with another equally unsupported story.

The conversation, continued

E Conversations / turns 11-20

04

Continue with the word bank on page 37.

- 11 **Marcus:** For the numerical wording, ninety minus eighty is ten percentage points. Would a twelve-point-five-percent improvement headline simply express the same observed difference another way?
- 12 **Priya:** It expresses ⁶ _____ against the eighty-percent reference rate. But improvement may still imply cause. Say a twelve-point-five-percent higher observed rate, and keep the nonrandom attendance qualification nearby.
- 13 **Marcus:** The business question is whether offering or requiring training would improve renewal for eligible customers. That is different from describing people who already chose to attend.
- 14 **Priya:** Yes. The ⁷ _____ question is what comparable customers would do without the intervention. We need a reviewed evaluation approach suited to that decision, not just a more persuasive headline.
- 15 **Marcus:** We may be able to add customer size and tenure. Would adjusting for those settle the concern, or would important differences still be missing?
- 16 **Priya:** A suitable ⁸ _____ may help with measured differences, but it does not automatically address unmeasured motivation or every source of bias. The method and assumptions need explicit review.
- 17 **Marcus:** And a mandatory program could reach people unlike the voluntary attendees. Even a stronger estimate in this group might not transfer unchanged to everyone.
- 18 **Priya:** That is a ⁹ _____ issue as well as a causal one. The population, intervention, and implementation matter when translating evidence into a wider operational recommendation.
- 19 **Marcus:** I will revise the slide to higher observed renewal among attendees, show both counts and rates, and recommend a better evaluation before claiming a guaranteed benefit.
- 20 **Priya:** Good. A reviewed ¹⁰ _____ can address the decision we actually face. We have kept the useful finding, corrected its interpretation, and made the next evidence step clear.

ANSWERS / EXPLANATIONS: PAGE 92

1. association | 2. self-selection | 3. causal effect | 4. Confounding | 5. unmeasured factor | 6. relative lift | 7. counterfactual | 8. adjustment | 9. generalizability | 10. evaluation design

A difference does not supply its cause

F Say it

In a voluntary program, 45 of 50 attendees renew and 80 of 100 non-attendees renew. Baseline engagement is unmeasured. Attendance was not randomized.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

ten / 80 / unestablished / 90

Analyst: "Attendee renewal is ¹ _____ percent."

Lead: "Non-attendee renewal is ² _____ percent."

Analyst: "The observed gap is ³ _____ percentage points."

Lead: "The causal effect remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 92

1. 90 | 2. 80 | 3. ten | 4. unestablished

Rehearse with a fixed script

1. Read turns 1-10. Stress observed association and voluntary attendance.
2. Switch roles for turns 11-20. Keep a possible confounder distinct from a proven explanation.
3. Check the transfer. Read both group rates without attributing the gap to training.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The join adds rows, not order value

A Read the situation

Explain row grain and join multiplicity, diagnose a duplicated aggregate, and reject a misleading distinct-value fix.

05

Analytics engineer Lina reviews a report with analyst Owen. The complete teaching dataset has two orders: O1 is \$100 and O2 is \$100. Payment rows are O1: \$60 and \$40; O2: \$100. Joining orders to payments on order ID creates three rows. Summing the repeated order amount then gives \$300, although total order value is \$200. `SUM(DISTINCT order_amount)` gives \$100 because the two legitimate orders share an amount. Lina proposes aggregating payments to one row per order before joining, with key and total checks.

PEOPLE / Lina: Analytics engineer | Owen: Reporting analyst

EXPLAIN THE ROW, THEN THE TOTAL

When a report suddenly grows after a join, avoid treating the larger figure as business growth or blaming a source team without checking the transformation. A small concrete example lets technical and business colleagues inspect the same logic and agree on a verifiable repair.

Briefing checks

Circle one answer in each item. Use the briefing.

1. How many rows result from the supplied join?

A Two

B Four

C One

D Three

2. What is the actual total order value?

A \$160

B \$200

C \$300

D \$100

3. Why does `SUM(DISTINCT order_amount)` give \$100 here?

A It deduplicates correctly by order ID.

B It adds all payment amounts.

C It restores the missing third order.

D Both legitimate orders have the same amount, so distinct values collapse them.

Answers and explanations: [page 93](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

SQL

Structured Query Language, used to query relational data.

IN USE review SQL logic

row grain

What one row represents in a dataset.

IN USE define the row grain

primary key

A non-null column or set of columns uniquely identifying each row.

IN USE validate the primary key

foreign key

A column or set of columns constrained to reference a key in a table.

IN USE check foreign-key relationships

join key

The field or fields used to match tables.

IN USE choose the join key

join cardinality

The relationship in match counts between joined entities.

IN USE inspect join cardinality

one-to-many relationship

One entity matching multiple related records.

IN USE model a one-to-many relationship

fanout

Multiplication of rows caused by multiple join matches.

IN USE detect join fanout

aggregation

Combining records into summary values.

IN USE control aggregation grain

pre-aggregation

Summarizing data before another transformation or join.

IN USE apply pre-aggregation

GROUP BY

SQL grouping of rows for aggregate calculations.

IN USE group by order ID

SUM

An aggregate that adds non-null numeric values.

IN USE check the SUM result

Precision check

The joined order amounts are 100, 100, and 100: O1 appears twice because it has two payments. The correct order total is 200. The payment amounts themselves are 60, 40, and 100, also totaling 200 for this dataset.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

DISTINCT

SQL removal of duplicate values or rows in its specified scope.

IN USE inspect **DISTINCT** semantics

COUNT

An aggregate that counts rows or non-null expressions.

IN USE distinguish **COUNT** variants

inner join

A join returning matching combinations from both sides.

IN USE inspect inner-join matches

left join

A join retaining left rows even without a right match.

IN USE preserve orders with a left join

null handling

Rules for processing missing or unknown values.

IN USE define null handling

filter

A condition selecting which records remain.

IN USE apply the filter

CTE

Common table expression, a named query expression.

IN USE use a CTE

data model

A representation of entities, relationships, and measures.

IN USE review the data model

fact table

A table storing business events or measurements at a defined grain.

IN USE define the fact-table grain

dimension table

A table holding descriptive attributes for analysis.

IN USE maintain dimension keys

reconciliation total

A reference total used to verify transformed output.

IN USE compare reconciliation totals

uniqueness test

A check that a defined key has no duplicates.

IN USE run a uniqueness test

Precision check

SUM(DISTINCT order_amount) removes duplicate numeric values, not duplicate order identities. It wrongly collapses the two \$100 orders to one amount. Restore the intended grain and verify keys, unmatched rows, and totals instead of using a convenient-looking shortcut.

Phrases you can actually use

C Notice the language / 1 of 2

ASK ABOUT GRAIN

"What does one row represent before the join?"

DESCRIBE THE RELATIONSHIP

"O1 has two payment rows."

SEPARATE DATA FROM GROWTH

"The extra row is not an extra order."

GIVE THE REFERENCE

"The order-level total is \$200."

NAME THE KEY

"The tables match on order ID."

EXPLAIN THE FANOUT

"The order row is repeated for each matching payment."

GIVE THE WRONG TOTAL

"Summing joined order amounts produces \$300."

REJECT THE SHORTCUT

"Distinct amounts do not mean distinct orders."

Row

Not always the same as a business entity.

Many

Indicates multiple matches, not necessarily bad source data.

Distinct

Ask distinct on which fields or values.

Phrases you can actually use

C Notice the language / 2 of 2

SHOW THE COUNTEREXAMPLE

"Both legitimate orders have an amount of \$100."

VERIFY THE KEY

"Check uniqueness of order ID in the summarized payments."

COMPARE TOTALS

"Reconcile the repaired output to the source order total."

KEEP NULL RULES EXPLICIT

"Define how orders without payments are represented."

PROPOSE THE REPAIR

"Aggregate payments to one row per order before joining."

PRESERVE UNMATCHED ORDERS

"The intended order population must survive the join."

SEPARATE MEASURES

"Order value and payment value are different measures."

CLOSE THE REVIEW

"Fix the grain and verify the result before republishing."

Sum

Depends on the grain of the rows being summed.

Left join

Preserves left-side presence but does not prevent fanout.

Fixed

Requires checks beyond making one displayed total look plausible.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which operation explains the \$300 result?

- A** Summing the actual 60, 40, and 100 payment amounts
- B** Adding the two original order amounts once each
- C** Counting distinct order IDs
- D** Summing order amounts after O1 is repeated for two payments

2. Why is summing distinct amounts not a valid repair?

- A** It changes O2 into a zero-value order.
- B** It merges different orders that happen to have equal values.
- C** It always preserves every order identity.
- D** It forces payment rows to become unique by ID.

3. Which proposed repair fits this report?

- A** Replace the inner join with a left join and sum the repeated order values.
- B** Summarize payments by order ID, then join at one row per order and verify totals.
- C** Keep the payment-detail join and sum distinct order amounts.
- D** Group payments by amount before joining to each order amount.

4. Which statement about a left join is accurate?

- A** It can preserve unmatched orders but still duplicate an order with multiple matches.
- B** It always guarantees one row per order.
- C** It prevents every duplicate amount.
- D** It discards every left-side order without payment.

Answers and explanations: [page 93](#).

The join adds rows, not order value

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

05

WORD + PHRASE BANK

one-to-many relationship / uniqueness test / join key / DISTINCT / aggregation / left join / reconciliation total / fanout / pre-aggregation / row grain

- 01 **Owen:** The report jumped from two hundred dollars of order value to three hundred after I added payment details. I thought the join would only add columns.
- 02 **Lina:** Check the ¹ _____. Before the join, each row represents one order. After matching O1 to two payment records, that order appears in two joined rows.
- 03 **Owen:** O1 is one hundred dollars with payments of sixty and forty. O2 is another one-hundred-dollar order with a single one-hundred-dollar payment.
- 04 **Lina:** That is a ² _____. The join produces three rows: two for O1 and one for O2. The third row is not evidence of another business order.
- 05 **Owen:** I summed the order amount after the join. Each resulting row displays one hundred, so the three displayed amounts add to three hundred.
- 06 **Lina:** That is join ³ _____. The payment rows are legitimate, but the order-level measure is repeated. We need to preserve its meaning instead of treating each joined occurrence as new value.
- 07 **Owen:** SUM DISTINCT brought it down to one hundred. That looks worse: we have two separate orders worth one hundred each. Have I removed the wrong duplicate?
- 08 **Lina:** Exactly. ⁴ _____ applies to the selected value here. Two different orders can both be worth one hundred dollars, so amount equality does not establish duplicate business identity.
- 09 **Owen:** The correct order total is two hundred. The payment amounts also total two hundred in this small dataset, but that does not make the measures identical.
- 10 **Lina:** Keep the ⁵ _____ tied to the intended entity. An order-value report and a payment-value report answer different questions, even when their totals happen to match in this example.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 46.

- 11 **Owen:** For the order report, I can summarize the payment side first so there is one payment-summary row for each order ID.
- 12 **Lina:** Yes, use ⁶ _____. Then verify that the payment summary has only one row per order. Joining two aligned grains avoids repeating the original order amount for every payment.
- 13 **Owen:** The grouping field should be order ID, not amount. Grouping by amount would combine the two orders again because they both have the same value.
- 14 **Lina:** Correct. The ⁷ _____ must preserve the intended relationship. Test the summary key and check that the matching logic uses the appropriate identifier on both sides.
- 15 **Owen:** This teaching dataset has payments for both orders. In the full report, an order without a payment should still appear if it belongs to the reporting population.
- 16 **Lina:** Then choose and test the appropriate ⁸ _____. It can preserve unmatched orders, but it would not by itself solve multiple matches. Population preservation and multiplicity are separate checks.
- 17 **Owen:** I will check both the two-order count and the two-hundred-dollar total. Could those totals look right even if one missing order offsets a duplicate?
- 18 **Lina:** Use the original ⁹ _____ as a reference and inspect differences. A plausible-looking grand total alone is not enough if missing orders and duplicated orders happen to offset each other.
- 19 **Owen:** My explanation will say the join repeated an order-level amount; it did not create revenue. I will withhold the inflated result while the transformation is corrected.
- 20 **Lina:** Add a ¹⁰ _____ for the summarized order key and document the grain. That makes the repair repeatable rather than relying on someone noticing another surprising total later.

ANSWERS / EXPLANATIONS: PAGE 94

1. row grain | 2. one-to-many relationship | 3. fanout | 4. DISTINCT | 5. aggregation | 6. pre-aggregation | 7. join key | 8. left join | 9. reconciliation total | 10. uniqueness test

Equal amounts can belong to different orders

F Say it

Order A is \$80 with two \$40 payments. Order B is \$80 with one \$80 payment. An order-to-payment join yields three rows. The report must count each order amount once.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

80 / identity / 160 / 240

Engineer: "The correct order total is \$¹ _____."

Analyst: "The joined sum of order amounts is \$² _____."

Engineer: "The sum of distinct amount values is \$³ _____."

Analyst: "The repair must preserve order⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 94

1. 160 | 2. 240 | 3. 80 | 4. identity

Rehearse with a fixed script

1. Read turns 1-10. Contrast an order identity with its amount.
2. Switch roles for turns 11-20. Stress one row per order and both reconciliation checks.
3. Check the transfer. Read the correct total, inflated sum, and distinct-value sum separately.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

A positive estimate is not the rollout rule

A Read the situation

Present an experiment estimate with its interval, practical threshold, and pre-agreed decision rule.

06

Experiment analyst Theo reviews a completed randomized test with product director Hana. The validated readout estimates a 1.2-percentage-point increase in completion, with a 95% confidence interval from -0.4 to +2.8 points. The pre-agreed rollout rule requires the interval lower bound to exceed +1.0 point, with guardrails passing. Guardrails passed, but the lower-bound condition did not. The slide currently shows only +1.2. The test has reached its planned endpoint; no unplanned extension or new rollout approval is authorized.

PEOPLE / Theo: Experiment analyst | Hana: Product director

DISTINGUISH A PROMISING SIGNAL FROM A PASSED RULE

A positive estimate can be worth discussing without being a win under the actual decision standard. Show the interval and threshold together. Correcting an overconfident slide should preserve the evidence, not replace optimism with an equally unsupported claim of no effect.

Briefing checks

Circle one answer in each item. Use the briefing.

1. What is the point estimate?

A -0.4 percentage points

C +1.2 percentage points

B +1.0 percentage point

D +2.8 percentage points

2. Did the lower-bound rule pass?

A No; -0.4 does not exceed +1.0.

C Yes; every positive estimate passes.

B Yes; the point estimate exceeds +1.0.

D No; guardrails failed.

3. What stage has the test reached?

A A completed rollout

C An unplanned first-day peek

B Its planned endpoint

D An automatically approved extension

Answers and explanations: [page 95](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

randomized experiment

A study allocating exposure through a random process.

IN USE interpret a randomized experiment

point estimate

A single value estimating an unknown quantity.

IN USE report the point estimate

interval estimate

A range estimating an unknown quantity under a method.

IN USE report an interval estimate

confidence interval

An interval from a method with specified long-run coverage.

IN USE interpret a confidence interval

confidence level

The nominal long-run coverage rate of the interval method.

IN USE state the confidence level

lower bound

The smaller endpoint of an interval.

IN USE compare the lower bound

upper bound

The larger endpoint of an interval.

IN USE report the upper bound

effect size

The magnitude of a difference or relationship.

IN USE assess effect size

no-change value

The parameter value representing no effect on the stated scale.

IN USE identify the no-change value

practical threshold

The effect boundary relevant to a business decision.

IN USE set a practical threshold

decision rule

A predefined condition governing a conclusion or action.

IN USE apply the decision rule

guardrail

A measure limiting unacceptable side effects.

IN USE check the guardrails

Precision check

The point estimate is +1.2 points, but the interval runs from -0.4 to +2.8. It includes zero and values below the practical threshold. Under the supplied rule, a passing guardrail cannot compensate for a failing lower bound.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

primary outcome

The main measure used to assess the intervention.

IN USE specify the primary outcome

planned endpoint

The pre-agreed point for completing the analysis or test.

IN USE respect the planned endpoint

statistical significance

A result meeting a specified statistical testing criterion.

IN USE qualify statistical significance

practical significance

Importance of an effect for a real decision.

IN USE assess practical significance

null hypothesis

A specified claim assessed by a statistical test.

IN USE state the null hypothesis

standard error

The standard deviation of an estimator's sampling distribution, usually estimated.

IN USE report the standard error

sampling variability

Variation in estimates across possible samples.

IN USE account for sampling variability

statistical power

Probability of detecting a specified effect under assumptions.

IN USE plan statistical power

minimum detectable effect

An effect size used in the study's power planning.

IN USE specify the minimum detectable effect

sequential analysis

A method allowing planned repeated statistical decisions.

IN USE use a valid sequential analysis

equivalence

Similarity within a predefined acceptable difference margin.

IN USE test equivalence appropriately

inconclusive result

Evidence insufficient for the specified conclusion.

IN USE explain an inconclusive result

06

Precision check

A 95% confidence method has a stated long-run coverage property under its assumptions; it does not make the realized fixed parameter 95% likely to lie in this particular interval. An interval including zero also does not prove equivalence or no effect.

Phrases you can actually use

C Notice the language / 1 of 2

REPORT THE CENTER

"The estimated effect is plus 1.2 percentage points."

NAME THE NO-CHANGE VALUE

"The interval includes zero."

STATE THE ACTUAL RULE

"The lower bound must exceed plus 1.0 point."

PRESERVE THE GUARDRAIL RESULT

"The guardrails passed."

SHOW THE UNCERTAINTY

"The 95% interval runs from minus 0.4 to plus 2.8 points."

KEEP THE SCALE EXPLICIT

"These are percentage points, not relative percentages."

APPLY THE RULE

"The lower-bound condition was not met."

SEPARATE THE CONDITIONS

"Passing guardrails does not replace the effect requirement."

06

Positive

Describes the estimate's sign, not guaranteed benefit.

Includes zero

Does not establish an exact zero effect.

95%

Describes the method's confidence level, not every possible uncertainty.

Phrases you can actually use

C Notice the language / 2 of 2

AVOID DECLARING A WINNER

"This readout does not authorize rollout under our rule."

AVOID DECLARING NO EFFECT

"It also does not prove the true effect is zero."

06

RESPECT THE ENDPOINT

"The planned test has ended."

REJECT OPPORTUNISTIC EXTENSION

"We cannot keep collecting until the result looks favorable."

DISCUSS NEXT WORK HONESTLY

"Any new study needs an explicitly reviewed design."

KEEP THE INTERVAL VISIBLE

"Place both bounds beside the estimate."

DISTINGUISH DECISION FROM INTEREST

"The signal may merit follow-up without passing this launch rule."

CLOSE THE READOUT

"Report the estimate, interval, guardrails, and unmet decision condition together."

Significant

Distinguish statistical criteria from business importance.

Failed rule

Not the same as proof that an intervention is useless.

More data

Needs a justified design, not a search for a preferred result.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which slide is most faithful to the readout?

- A** +1.2 points; 95% interval -0.4 to +2.8; rollout approved because guardrails pass
- B** +1.2% relative lift; 95% interval -0.4 to +2.8 points; criterion unmet
- C** +1.2 points; 95% interval +0.4 to +2.8; guardrails pass; criterion unmet
- D** +1.2 points; 95% interval -0.4 to +2.8; guardrails pass; rollout criterion unmet

2. Why does the +1.2 estimate not satisfy the rule?

- A** Every positive effect must be ignored.
- B** Guardrails always determine effect size.
- C** The rule uses the interval lower bound, which is -0.4.
- D** The rule requires the upper bound to be negative.

3. What would be an unsupported next step?

- A** Report that the decision condition was unmet.
- B** Review a properly designed follow-up study.
- C** Keep the interval visible on the slide.
- D** Extend the finished test repeatedly until the original method produces a favorable result.

4. Which statement about zero is correct?

- A** Inclusion proves the treatment and control are identical.
- B** Inclusion means no further study can be useful.
- C** Inclusion makes the interval a probability distribution over customers.
- D** Its inclusion does not prove either exact zero effect or equivalence.

Answers and explanations: [page 95](#).

A positive estimate is not the rollout rule

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

06

WORD + PHRASE BANK

practical threshold / inconclusive result / confidence interval / planned endpoint / practical significance / no-change value / decision rule / lower bound / sequential analysis / point estimate

- 01 **Hana:** The slide says plus one point two, which looks promising. Can we call the test a win and move to rollout now that it has finished?
- 02 **Theo:** Show the ¹ _____ with the interval. The estimated increase is one point two percentage points, but the ninety-five-percent interval extends from minus zero point four to plus two point eight.
- 03 **Hana:** That range includes a small negative effect and zero as well as useful positive outcomes. The current slide hides those possibilities by showing only the center.
- 04 **Theo:** Correct. The ² _____ belongs beside the estimate. It prevents us from presenting a promising central value as if the uncertainty were negligible or entirely favorable.
- 05 **Hana:** Remind me of the actual rollout rule. I remember that we set a one-percentage-point threshold and also required the guardrails to pass.
- 06 **Theo:** The ³ _____ must exceed plus one point zero, with guardrails passing. Our lower endpoint is minus zero point four, so that condition was not met.
- 07 **Hana:** The guardrails did pass, which is good news. But that is a separate condition, not a substitute for the evidence requirement on completion improvement.
- 08 **Theo:** Exactly. Apply the whole ⁴ _____. We can acknowledge the guardrail result while saying the test did not authorize rollout under the standard agreed before seeing these results.
- 09 **Hana:** Then should the slide say no effect? Zero is inside the interval, but so are positive effects. I do not want to replace one overclaim with another.
- 10 **Theo:** No. The ⁵ _____ is included, but useful positive effects are included too. The readout does not establish exact zero or show that the two experiences are equivalent.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 55.

- 11 **Hana:** Then the decision is clearer than the underlying effect. We know the supplied rollout rule failed, while the magnitude and even direction remain uncertain.
- 12 **Theo:** That is an ⁶ _____ for the desired benefit claim, not permission to hide the estimate. We should report what the evidence supports and what decision it does not support.
- 13 **Hana:** Could we extend it until the lower bound clears one point? Or would that change the rule after we have already seen the result?
- 14 **Theo:** We reached the ⁷ _____. Continuing until the result looks favorable changes the statistical behavior. There is no authorized extension or approved repeated-decision method in the supplied plan.
- 15 **Hana:** A new study might still be justified, but it would need a clear design and rationale rather than an attempt to rescue the original headline.
- 16 **Theo:** Yes. An appropriate ⁸ _____ can support planned repeated decisions, but that is not what was specified here. Any new evaluation needs its own reviewed assumptions and rules.
- 17 **Hana:** I will also keep the units visible. Plus one point two means percentage points, not a one-point-two-percent relative increase from an unstated baseline.
- 18 **Theo:** Good. The ⁹ _____ should be displayed on the same scale. Statistical uncertainty and business value are related in the decision, but they are not interchangeable labels.
- 19 **Hana:** My revised readout will show the center, both bounds, passing guardrails, and the unmet rollout condition. No launch approval or no-effect conclusion will be implied.
- 20 **Theo:** That preserves ¹⁰ _____ as a separate question. The signal may merit well-designed follow-up, but this completed test has not passed the agreed rule for rollout.

06

ANSWERS / EXPLANATIONS: PAGE 96

1. point estimate | 2. confidence interval | 3. lower bound | 4. decision rule | 5. no-change value | 6. inconclusive result | 7. planned endpoint | 8. sequential analysis | 9. practical threshold | 10. practical significance

Read the endpoint named in the rule

F Say it

An experiment estimates +0.8 points with a 95% interval from -0.2 to +1.8. The rule requires the lower bound to exceed +0.5, and guardrails pass. The planned test has ended.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

0.5 / 0.2 / unmet / 0.8

Analyst: "The point estimate is plus ¹ _____ points."

Director: "The lower bound is minus ² _____ points."

Analyst: "The required boundary is plus ³ _____ points."

Director: "The rollout condition is ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 96

1. 0.8 | 2. 0.2 | 3. 0.5 | 4. unmet

Rehearse with a fixed script

1. Read turns 1-10. Say minus and plus clearly for both interval endpoints.
2. Switch roles for turns 11-20. Contrast an unmet rollout rule with proof of no effect.
3. Check the transfer. Stress the lower bound rather than the positive point estimate.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

The analysis needs weekly counts, not customer names

A Read the situation

Specify a purpose-limited data request, distinguish pseudonymization from anonymity, and preserve approval and retention controls.

Analyst Sara asks data steward Malik for a full customer export to compare weekly late-resolution rates by product tier. The supplied analysis can use week, tier, eligible-case count, and late-case count; names, emails, phone numbers, and free-text comments are unnecessary. A team proposal uses a reviewed aggregate table in the approved workspace with 30-day access, subject to owner approval and disclosure review. Approval is pending. Small groups still require review. Replacing names with stable IDs would not automatically make row-level data anonymous or remove controls.

PEOPLE / Sara: Data analyst | Malik: Data steward

07

TIE EACH FIELD TO THE QUESTION

A constructive data-access discussion explains what information the analysis requires and offers a usable alternative. Avoid assuming that analysts need everything or that every aggregate is harmless. Keep the purpose, approval, audience, and handling conditions visible in the same request.

Briefing checks

Circle one answer in each item. Use the briefing.

1. Which fields directly support the supplied analysis?

A Passwords and private payment credentials

B Week, tier, eligible-case count, and late-case count

C Every customer's name and phone number

D Full free-text comments with unrelated details

2. What is the current approval status?

A Pending

B Automatic because the data is aggregated

C Granted by the analyst's request

D Permanent approval for every future purpose

3. What does replacing names with stable IDs establish?

A Every row can be publicly released.

B It does not automatically establish anonymity.

C All access controls become unnecessary.

D Re-identification is impossible in every context.

Answers and explanations: [page 97](#).

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

data governance

Responsibilities and rules for managing data use.

IN USE establish data governance

data steward

A role supporting defined data quality and handling responsibilities.

IN USE consult the data steward

data owner

The role accountable for specified data decisions.

IN USE obtain data-owner approval

purpose limitation

Restricting data use to defined appropriate purposes.

IN USE document the analysis purpose

data minimization

Limiting data to what the stated purpose needs.

IN USE apply data minimization

personal data

Information relating to an identifiable person under the applicable rules.

IN USE assess personal-data handling

direct identifier

A field that directly identifies a person.

IN USE remove unnecessary direct identifiers

indirect identifier

A field that may identify someone when combined with other information.

IN USE review indirect identifiers

pseudonymization

Replacing identifying details while retaining possible linkage.

IN USE assess pseudonymization safeguards

anonymization

Processing intended to prevent identification under the relevant standard.

IN USE evaluate anonymization

re-identification risk

The possibility of connecting data back to an individual.

IN USE assess re-identification risk

aggregation

Combining records into group-level summaries.

IN USE review aggregation level

Precision check

The rate needs late-case count divided by eligible-case count within each week and tier. Names and contact details do not support that supplied calculation. Data minimization means enough relevant data for the purpose, not an unusably stripped dataset.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

small cell

A reported group with few contributing observations.

IN USE review small cells

disclosure review

Assessment of risks from releasing or sharing information.

IN USE conduct disclosure review

access scope

The specific data and actions a permission covers.

IN USE limit access scope

retention period

The duration for which data is kept under defined rules.

IN USE document the retention period

access expiry

The point when an access permission ends.

IN USE enforce access expiry

approved workspace

An environment authorized for the stated data work.

IN USE use the approved workspace

data lineage

The traceable path of data through sources and transformations.

IN USE document data lineage

data catalog

An organized inventory of datasets and their metadata.

IN USE consult the data catalog

processing purpose

The defined reason for collecting or using data.

IN USE state the processing purpose

permitted audience

The people authorized to receive a result or dataset.

IN USE define the permitted audience

export restriction

A limit on moving data outside an approved environment.

IN USE enforce export restrictions

secondary use

A use beyond the original stated purpose.

IN USE review proposed secondary use

Precision check

Thirty-day access is the proposed local permission duration, not a universal legal retention rule. Access expiry and data deletion are separate matters. Aggregation or stable-ID substitution does not automatically remove disclosure risks or approval requirements.

Phrases you can actually use

C Notice the language / 1 of 2

DEFINE THE PURPOSE

"We are comparing weekly late-resolution rates by tier."

LIMIT IDENTIFIERS

"Names and contact details are unnecessary for this question."

PRESERVE THE DENOMINATOR

"Include eligible-case counts as well as late-case counts."

STATE THE ENVIRONMENT

"The work belongs in the approved workspace."

LINK FIELDS TO THE TASK

"The calculation needs counts by week and tier."

OFFER AN ALTERNATIVE

"Use the reviewed aggregate table."

KEEP APPROVAL VISIBLE

"Owner approval is still pending."

BOUND ACCESS

"The proposed access duration is thirty days."

Necessary

Explain the connection between each field and the question.

Aggregate

A format, not an automatic privacy determination.

Anonymous

A conclusion requiring the applicable assessment.

Phrases you can actually use

C Notice the language / 2 of 2

SEPARATE DURATION RULES

"Access expiry is not automatically the data-retention rule."

AVOID AN ANONYMITY CLAIM

"Stable IDs do not automatically make the records anonymous."

KEEP EXPORTS GOVERNED

"Do not assume approval permits an external export."

RECORD THE SOURCE

"Keep the source and transformation lineage."

CHECK GROUP DISCLOSURE

"Small groups still need disclosure review."

CONTROL THE AUDIENCE

"Confirm who may receive the output."

ADDRESS NEW PURPOSES

"A different question may need a new review."

CLOSE THE REQUEST

"Submit the purpose, four fields, workspace, duration, and review conditions."

Pseudonymous

May still allow linkage and require protection.

Thirty days

A case-specific access proposal, not a legal rule.

Approved

State exactly which purpose, audience, and actions are covered.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which request best matches the supplied need?

- A** Reviewed weekly tier-level counts in the approved workspace, with pending approvals and 30-day access
- B** A full customer export with unlimited future use
- C** Names and phone numbers without case counts
- D** Public release of stable-ID records without review

07

2. Why include eligible-case count?

- A** It provides the denominator for each rate.
- B** It identifies every customer's phone number.
- C** It proves all small groups are anonymous.
- D** It replaces owner approval.

3. Which statement about aggregation is accurate?

- A** It can reduce detail but still needs the stated disclosure review.
- B** Every aggregate is automatically safe to publish.
- C** It always preserves every individual identifier.
- D** It removes the need for an analysis purpose.

4. What does 30-day access specify?

- A** Permanent permission after the first month
- B** Approval already granted to all employees
- C** The proposed duration of permission in this case
- D** A universal legal deletion deadline

Answers and explanations: [page 97](#).

The analysis needs weekly counts, not customer names

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

access scope / Access expiry / data minimization / pseudonymization / secondary use / processing purpose / disclosure review / aggregation / data lineage / permitted audience

07

- 01 **Sara:** I requested the full customer export for weekly late-resolution rates by product tier. I thought extra fields would give us flexibility.
- 02 **Malik:** Start with the ¹ _____. The supplied question needs week, tier, eligible-case count, and late-case count. Which part requires names, contact details, or unrestricted free-text comments?
- 03 **Sara:** Those extra fields do not enter the rate. I need late cases divided by eligible cases within each week and tier.
- 04 **Malik:** Then ² _____ supports a narrower request. We should retain what makes the analysis valid, including its denominator, without adding unrelated personal fields simply because they might be interesting later.
- 05 **Sara:** An aggregate table would work for this comparison. I do not need individual rows if the counts use the agreed definitions and can be traced to their source.
- 06 **Malik:** That ³ _____ is a useful alternative, subject to review. We still need to check the output, especially where a group is small enough to raise identification or disclosure concerns.
- 07 **Sara:** So a small tier in one week should not automatically be published just because its row contains counts rather than names. The audience and context still matter.
- 08 **Malik:** Correct. A ⁴ _____ remains part of the proposal. Do not treat a change in table shape as an automatic conclusion that all privacy and handling conditions have disappeared.
- 09 **Sara:** If someone later asks for individual records, could replacing customer names with stable IDs make that request anonymous and remove the approval requirement?
- 10 **Malik:** That is usually a ⁵ _____ question, not an automatic anonymity claim. Stable identifiers can still support linkage, and the applicable assessment and access controls must remain explicit.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 64.

- 11 **Sara:** For this task, I will stay with the reviewed aggregate fields. The proposal also specifies the approved workspace rather than an unrestricted download to any location.
- 12 **Malik:** Keep that in the ⁶ _____. Owner approval is still pending. A technically convenient format does not authorize an export or make the request approved before the decision is recorded.
- 13 **Sara:** Does thirty-day access also mean deleting the data on day thirty? I need to distinguish the permission expiry from the actual retention rule.
- 14 **Malik:** Exactly. ⁷ _____ and data retention are separate questions. The actual retention and deletion arrangements need their own approved basis rather than being inferred from this access duration.
- 15 **Sara:** I will also say who needs the results. The service manager needs the reviewed weekly comparison, but that does not imply that anyone may receive the underlying data.
- 16 **Malik:** Define the ⁸ _____ and the output route. The conditions should remain attached to the analysis even after the calculations are complete and somebody wants to reuse a chart.
- 17 **Sara:** If we later need customer-level linkage, I will bring a new purpose and field list. This request should not become permission for unrelated analysis.
- 18 **Malik:** That would be a ⁹ _____ review. We can assess the new need on its facts, but the current weekly-rate purpose should not become a blanket authorization for every future analysis.
- 19 **Sara:** My request will list the four fields, rate, workspace, thirty-day access, audience, and required reviews. Approval remains pending.
- 20 **Malik:** Include the ¹⁰ _____ for the aggregate counts as well. That preserves both analytical trust and the handling boundary while giving you the information the actual question needs.

07

ANSWERS / EXPLANATIONS: PAGE 98

1. processing purpose | 2. data minimization | 3. aggregation | 4. disclosure review | 5. pseudonymization | 6. access scope | 7. Access expiry | 8. permitted audience | 9. secondary use | 10. data lineage

A denominator is necessary; a phone number is not

F Say it

An analyst needs monthly complaint rates by region. The approved design uses month, region, complaint count, and eligible-order count. Names and phone numbers are unnecessary. Access approval is pending; proposed access lasts 14 days.

07

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

count / pending / unnecessary / 14

Analyst: "The rate denominator is the eligible-order ¹ _____."

Steward: "Phone numbers are ² _____ for this question."

Analyst: "The proposed access period is ³ _____ days."

Steward: "Approval remains ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 98

1. count | 2. unnecessary | 3. 14 | 4. pending

Rehearse with a fixed script

1. Read turns 1-10. Stress the four needed fields and remaining disclosure review.
2. Switch roles for turns 11-20. Keep access expiry distinct from retention.
3. Check the transfer. Read the fourteen-day proposal without implying approval already exists.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Five minutes to authorize the right investigation

A Read the situation

Lead a short briefing with a bounded action, quantified evidence, explicit limits, and a review point.

Analyst Oren has five minutes with operations director Claudia. Two complete weeks use the same ticket definition: issue Q appears in 20 of 1,000 tickets in week one and 80 of 1,000 in week two. Seventy of the later 80 issue-Q tickets are mobile-related; the total mobile-ticket denominator is not supplied. A release occurred between weeks, but cause is unestablished. Oren requests a two-day investigation led by the service analyst, with a readout afterward. Approval is for investigation only, not a production change.

PEOPLE / Oren: Data analyst | Claudia: Operations director

08

MAKE UNCERTAINTY ACTIONABLE

A concise recommendation does not need an invented root cause. State the observed change, why it merits attention, and the smallest defined evidence step that supports the next decision. Acknowledge what a concentration of tickets does not tell you about the wider user population.

Briefing checks

Circle one answer in each item. Use the briefing.

1. How did the issue-Q ticket rate change?

A From 8% to 2%, a decline

B From 2% to 8%, a six-percentage-point rise

C From 20% to 80%, a sixty-point rise

D From 2% to 8%, a six-percent relative rise

2. What does the mobile count establish?

A The mobile failure rate is proven higher than desktop's.

B Seventy of the eighty later issue-Q tickets are mobile-related.

C 87.5% of every mobile user experienced the issue.

D Every mobile ticket concerns Q.

3. What authorization is requested?

A A two-day investigation with a subsequent readout

B An immediate unreviewed production change

C A declaration that the release caused Q

D Permanent access to all customer records

Answers and explanations: page 99.

The vocabulary that does the work

B Find the words / 1 of 2

Read the definitions and say the common word combinations aloud.

insight

A decision-relevant interpretation supported by evidence.

IN USE state the insight

recommendation

A proposed action with a reasoned basis.

IN USE frame the recommendation

decision ask

The precise choice requested from the audience.

IN USE lead with the decision ask

headline finding

The main evidence statement in a short briefing.

IN USE sharpen the headline finding

supporting evidence

Facts that substantiate a conclusion or recommendation.

IN USE present supporting evidence

issue category

A defined class used to group reported problems.

IN USE validate the issue category

ticket mix

The composition of support records across categories or groups.

IN USE analyze the ticket mix

segment concentration

A large share of observations within a particular subgroup.

IN USE report segment concentration

base rate

The underlying frequency relative to an eligible population.

IN USE check the base rate

denominator gap

Missing information needed to compute the relevant rate.

IN USE disclose the denominator gap

descriptive finding

A statement about observed data without a causal conclusion.

IN USE distinguish descriptive findings

causal hypothesis

A proposed explanation of what produced an outcome.

IN USE test the causal hypothesis

Precision check

Issue Q rises from 2% to 8% of tickets: +6 percentage points, four times the earlier rate, or a 300% relative increase. The later mobile share is $70/80 = 87.5\%$ of issue-Q tickets, not of mobile users.

The vocabulary that does the work

B Find the words / 2 of 2

Read the definitions and say the common word combinations aloud.

release timing

The timing of a product or system change.

IN USE compare release timing

diagnostic follow-up

Targeted analysis to investigate an observed issue.

IN USE scope diagnostic follow-up

investigation owner

The role accountable for the defined inquiry.

IN USE assign the investigation owner

time box

A specified limit on the duration of work.

IN USE set a time box

readout

A presentation of findings and their implications.

IN USE schedule the readout

action boundary

The limit of what a decision authorizes.

IN USE state the action boundary

production change

A modification to a live operating system or service.

IN USE approve a production change

decision log

A record of choices, reasons, and responsible people.

IN USE update the decision log

priority criterion

A stated basis for ordering work.

IN USE apply the priority criterion

evidence threshold

The support required for a specified conclusion or next action.

IN USE define the evidence threshold

leading question

A question that steers toward a preferred answer.

IN USE avoid leading questions

alternative explanation

Another plausible account of an observed pattern.

IN USE examine alternative explanations

Precision check

A release between the two weeks is a possible lead, not a demonstrated cause. Investigation approval does not authorize a production change. The two-day limit creates a readout point, not a guarantee that a root cause or fix will be found.

Phrases you can actually use

C Notice the language / 1 of 2

OPEN WITH THE ACTION

"I recommend a two-day investigation of issue Q."

NAME THE UNIT

"Those rates describe tickets, not unique users."

SHOW THE COUNTS

"The counts are 20 of 1,000 and 80 of 1,000."

LIMIT THE SEGMENT CLAIM

"We do not have the total mobile-ticket denominator."

STATE THE EVIDENCE

"The ticket rate rose from 2% to 8%."

GIVE THE CHANGE

"That is a six-percentage-point increase."

IDENTIFY THE CONCENTRATION

"Seventy of the later eighty issue tickets are mobile-related."

KEEP CAUSE OPEN

"The release timing is a lead, not proof of cause."

08

Rose

State the measure, unit, and comparison periods.

Four times

Means a 300% increase over the original, not 400% growth.

Mostly mobile

Describes case composition, not a denominator-based failure rate.

Phrases you can actually use

C Notice the language / 2 of 2

PROPOSE THE FOCUSED CHECK

"Examine affected mobile paths and classification consistency."

SET THE DURATION

"The proposed time box is two days."

PRESERVE THE BOUNDARY

"This request does not authorize a production change."

INVITE THE ACTUAL DECISION

"Please approve or decline that investigation scope."

NAME THE OWNER

"The service analyst would lead the investigation."

DEFINE THE OUTPUT

"Return with evidence, remaining uncertainty, and a next-step recommendation."

AVOID A DISCOVERY PROMISE

"The readout date is not a guaranteed fix date."

CLOSE THE BRIEFING

"The evidence supports investigating now without pretending the cause is settled."

08

After

Describes sequence without proving because of.

Two days

Bounds the work, not the certainty of its outcome.

Approve

Keep investigation authority separate from change authority.

Choose the precise message

D Check your understanding

Circle one best answer for each item. Use the stated purpose and case facts, not the option that merely sounds most polite.

1. Which opening best uses the five-minute meeting?

- A Approve a two-day investigation of Q; its ticket rate rose from 2% to 8%.
- B Here are ten unrelated tables with no action request.
- C The release definitely broke every mobile session.
- D We should change production immediately because the chart rose.

2. What is the relative increase from 2% to 8%?

- A 400%
- B 87.5%
- C 300%
- D 6%

3. Why is 70/80 not a mobile failure rate?

- A Seventy divided by eighty cannot be calculated.
- B Every issue ticket is a unique mobile user by definition.
- C The earlier week had no mobile users.
- D Its denominator is issue-Q tickets, not all relevant mobile activity.

4. What does approval of this request authorize?

- A Any immediate production change the analyst prefers
- B A public claim that the release caused the issue
- C An indefinite project without review
- D The defined two-day investigation and readout

Answers and explanations: [page 99](#).

Five minutes to authorize the right investigation

E Conversations / turns 1-10

Complete the numbered gaps. Use each bank entry once, without changing its form. Cover the answers at the end until you finish.

WORD + PHRASE BANK

headline finding / decision ask / denominator gap / time box / segment concentration / recommendation / supporting evidence / action boundary / investigation owner / causal hypothesis

08

- 01 **Claudia:** We have five minutes and ten findings. Which decision needs my attention now, and what evidence supports it?
- 02 **Oren:** My ¹ _____ is a two-day investigation of issue Q. Its share of tickets rose from two percent to eight percent across two complete weeks using the same classification.
- 03 **Claudia:** Give me the actual counts as well. I want to know whether the percentage changed because the ticket base became much smaller.
- 04 **Oren:** The ² _____ is twenty of one thousand tickets, then eighty of one thousand. The bases are equal, so the rate increase is not caused by a smaller denominator here.
- 05 **Claudia:** That is six percentage points, and four times the earlier rate. We should not call it six-percent growth or accidentally present ticket counts as unique customers.
- 06 **Oren:** Correct. The ³ _____ is a rise in the issue's ticket rate, not proof of how many distinct people were affected or why the issue increased.
- 07 **Claudia:** Seventy of the eighty are mobile-related. Can I say mobile fails more often than desktop, or are we missing the denominators for that comparison?
- 08 **Oren:** No. That is ⁴ _____ within the issue tickets. We lack the total mobile-ticket denominator, so we cannot calculate the relevant mobile rate or compare it fairly with desktop.
- 09 **Claudia:** But the concentration still gives the investigator a useful place to start. We can inspect the mobile paths without overstating what seventy out of eighty means.
- 10 **Oren:** Exactly. Disclose the ⁵ _____ and keep the descriptive finding. Seventy divided by eighty is eighty-seven point five percent of these issue tickets, not of mobile users.

The conversation, continued

E Conversations / turns 11-20

Continue with the word bank on page 73.

- 11 **Claudia:** A release occurred between the two weeks. That timing sounds relevant, although the current report does not show whether the release actually caused the increase.
- 12 **Oren:** Treat it as a ⁶ _____. The investigation should examine that lead and alternatives, including whether classification or affected paths changed. The sequence alone does not establish cause.
- 13 **Claudia:** Who would lead the work? I want a named role and a defined output, not an open request for somebody to analyze the problem someday.
- 14 **Oren:** The service analyst is the proposed ⁷ _____. The output is a readout of evidence, remaining uncertainty, and a recommended next step, rather than an unsupported promise of a fix.
- 15 **Claudia:** Two days for an evidence readout, not a guaranteed root cause. Please make that explicit in the invitation before people start expecting a fix.
- 16 **Oren:** It is a ⁸ _____, not a certainty deadline. We should report what the inquiry establishes and what remains unresolved at the readout, even if the original hypothesis is not supported.
- 17 **Claudia:** Does approving this request also allow the analyst to change production if a likely cause appears? I want that distinction written clearly before I decide.
- 18 **Oren:** The ⁹ _____ is investigation only. Any production change would need its separate evidence, review, and authority. We are not asking this meeting to bypass that process.
- 19 **Claudia:** Then the decision is whether to allocate two days to a documented service issue, with the mobile concentration as a lead and the cause still open.
- 20 **Oren:** Yes. The ¹⁰ _____ is approval of that scope. I will record your decision and preserve the ticket units, missing denominator, and causal limitation in the briefing.

ANSWERS / EXPLANATIONS: PAGE 100

1. recommendation | 2. supporting evidence | 3. headline finding | 4. segment concentration | 5. denominator gap | 6. causal hypothesis | 7. investigation owner | 8. time box | 9. action boundary | 10. decision ask

A concentration is a lead, not a rate

F Say it

Issue R appears in 10 of 500 tickets, then 30 of 500. Twenty-four of the later 30 issue tickets concern mobile. The total mobile base is unknown. The request is a one-day investigation, not a live change.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

six / four / investigation / 80

Analyst: "The later issue rate is ¹ _____ percent."

Director: "The increase is ² _____ percentage points."

Analyst: "Mobile represents ³ _____ percent of later issue tickets."

Director: "Approval would authorize an ⁴ _____."

ANSWERS / EXPLANATIONS: PAGE 100

1. six | 2. four | 3. 80 | 4. investigation

Rehearse with a fixed script

1. Read turns 1-10. Keep issue-ticket composition distinct from a mobile failure rate.
2. Switch roles for turns 11-20. Stress investigation only and the two-day readout.
3. Check the transfer. Read the 80% composition result with the issue-ticket denominator.

WORKING ALONE

Read both roles, pausing for two seconds when the speaker changes. Repeat with the bank covered, then check any missed expressions.

Both segments improve, but the overall rate falls

E More workplace conversations / 1-10

Explain weighted rates and changing composition without inventing a causal explanation.

Two periods each contain 1,000 eligible visits. Segment A changes from 10 conversions in 100 visits to 108 in 900. Segment B changes from 270 in 900 to 32 in 100. Definitions are unchanged. The team must explain segment improvement alongside an overall decline.

PEOPLE / Nadia: BI analyst | Paul: Commercial director

09

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

mix shift / causal effect / denominator / reference weights / standardization / percentage points

- 01 **Paul:** Your segment chart shows improvement in both groups, but the headline says conversion fell. Have the charts used different data?
- 02 **Nadia:** They use the same records. Start with each ¹ _____. A changes from one hundred to nine hundred visits; B changes from nine hundred to one hundred.
- 03 **Paul:** A goes from ten out of one hundred to one hundred eight out of nine hundred. That is ten percent to twelve percent.
- 04 **Nadia:** Yes, up two ² _____. B goes from thirty percent to thirty-two percent. Both segment-level comparisons are favorable.
- 05 **Paul:** The first total is ten plus two hundred seventy conversions: two hundred eighty out of one thousand, or twenty-eight percent.
- 06 **Nadia:** And the later total is one hundred eight plus thirty-two: one hundred forty out of one thousand, or fourteen percent. Those calculations reconcile.
- 07 **Paul:** So the later period has many more visits in the lower-converting segment. Its better rate does not offset that change in composition.
- 08 **Nadia:** Exactly. This ³ _____ explains why the aggregate comparison moves differently from the segment comparisons. An unweighted average would answer another question.
- 09 **Paul:** Could we just show the average of the two segment rates? That would be twenty percent before and twenty-two percent after.
- 10 **Nadia:** Only with an explicit equal-weight interpretation. It would not be the actual share of these visits that converted, because the segment sizes are not equal.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 76.

- 11 **Paul:** For a like-for-like composition comparison, could we apply the earlier visit proportions to the later segment rates?
- 12 **Nadia:** Yes, if we label the calculation. Use the earlier ⁴ _____: ten percent for A and ninety percent for B.
- 13 **Paul:** That gives point one times twelve percent plus point nine times thirty-two percent. The result is thirty percent, not the observed fourteen percent.
- 14 **Nadia:** Correct. It shows what that weighted calculation produces using the earlier mix. Keep the observed result alongside it; do not replace actual performance with the adjusted figure.
- 15 **Paul:** The commercial team might call thirty percent the performance we would certainly get if we changed our traffic allocation back.
- 16 **Nadia:** That overstates the ⁵ _____. It is a descriptive comparison, not proof that a future allocation change would preserve those rates.
- 17 **Paul:** Then we still need evidence about why the mix changed and whether different visitors within each segment would behave similarly.
- 18 **Nadia:** Yes. We have not estimated a ⁶ _____ of reallocating traffic. The arithmetic explains the apparent contradiction without settling the intervention decision.
- 19 **Paul:** I will show actual conversion, segment rates, and segment sizes together. The standardized comparison can be separate and clearly labeled.
- 20 **Nadia:** That lets the audience see both the real aggregate decline and the improvement within each segment, without choosing whichever chart tells the preferred story.

09

ANSWERS / EXPLANATIONS: PAGE 101

1. denominator | 2. percentage points | 3. mix shift | 4. reference weights | 5. standardization | 6. causal effect

An equal-weight average hides the actual visit mix

F Say it in a new situation

Earlier: A converts 20 of 100 visits; B converts 180 of 300. Later: A converts 90 of 300; B converts 70 of 100. Definitions match. Each period has 400 visits.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

composition / 50 / ten / 40

09

Analyst: The earlier overall rate is ¹ _____ percent.

Director: The later overall rate is ² _____ percent.

Analyst: Each segment improved by ³ _____ percentage points.

Director: Show the segment sizes so the changing ⁴ _____ is visible.

ANSWERS / EXPLANATIONS: PAGE 101

1. 50 | 2. 40 | 3. ten | 4. composition

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The customer moved regions. Did the old sale move too?

E More workplace conversations / 1-10

Agree historical attribution and distinguish business identity from a versioned dimension record.

Customer C17 belonged to North in February and moved to South effective March 1. Its February order is \$500; its April order is \$700. A current-region join assigns both to South. The requested report must use the region in effect on each order date.

PEOPLE / Iris: Analytics engineer | Mateo: Sales operations analyst

Use each bank entry once, without changing its form. Answers follow turn 20.

10

WORD + PHRASE BANK

current record / historical attribution / time-based lookup / restatement / surrogate key / Type 2 dimension

- 01 **Mateo:** South has gained five hundred dollars in February, even though nobody entered a new order. Could the customer-region update explain it?
- 02 **Iris:** Yes. The report joins both orders to the ¹ _____. It now says South, so the February order moves with the customer in that view.
- 03 **Mateo:** Our requested view is region at the time of sale. February should remain North and April should be South.
- 04 **Iris:** Then we need a ² _____ rule. Neither view should be labeled simply sales by region without telling readers which basis it uses.
- 05 **Mateo:** We have the approved March first effective date. Does the warehouse need two customer records, or would that duplicate the customer?
- 06 **Iris:** Two versions can represent one business customer. A ³ _____ preserves the region history instead of overwriting every previous state.
- 07 **Mateo:** C17 remains the business identifier in both versions. How do we distinguish the rows if that identifier is no longer unique in the history table?
- 08 **Iris:** Each version gets a ⁴ _____. Facts can then reference the correct version while we still group those versions under the same customer when needed.
- 09 **Mateo:** For the boundary, the North version ends when the South version starts. We must avoid matching the March first order to both.
- 10 **Iris:** Exactly. Define the effective intervals consistently. In this model the start is inclusive and the end exclusive, so March first belongs to South only.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 79.

- 11 **Mateo:** What does the loading process use to assign the version to each order? A current flag would give the same wrong answer again.
- 12 **Iris:** Use a ⁵ _____ against the effective interval for C17. Then test the boundary date, missing intervals, and overlapping versions.
- 13 **Mateo:** The expected result for these two orders is North five hundred, South seven hundred, total twelve hundred. The total alone would not expose the original error.
- 14 **Iris:** Right. The current-region report also totals twelve hundred. Check the regional allocation as well as the grand total, and retain both labeled views if both are required.
- 15 **Mateo:** Suppose the source team later corrects the move date. We should not silently replace an already issued regional report without a record.
- 16 **Iris:** Follow the agreed ⁶ _____ policy. Record the corrected source evidence, affected periods, and what changed in the published figures.
- 17 **Mateo:** Could we infer the old region from today's address if the earlier version is missing? That would make the chart complete.
- 18 **Iris:** Not without supporting evidence. Flag the unresolved history and route it to the source owner. A complete-looking chart is not a reason to invent an earlier state.
- 19 **Mateo:** I will confirm the requested historical basis with Sales, then check the two orders and the March first boundary in the revised output.
- 20 **Iris:** I will preserve C17's business identity, assign the appropriate version keys, and document the interval rule so future refreshes do not silently move old sales.

10

ANSWERS / EXPLANATIONS: PAGE 102

1. current record | 2. historical attribution | 3. Type 2 dimension | 4. surrogate key | 5. time-based lookup | 6. restatement

The grand total passes while the regional split is wrong

F Say it in a new situation

Customer D8 moves from East to West effective May 1. It has a \$300 April order and a \$400 June order. The report requires order-date region, not current region.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

keys / West / East / 700

10

Analyst: The April order belongs to ¹ _____.

Engineer: The June order belongs to ² _____.

Analyst: The unchanged grand total is ³ _____ dollars.

Engineer: Verify the effective-date intervals and the selected version
⁴ _____.

ANSWERS / EXPLANATIONS: PAGE 102

1. East | 2. West | 3. 700 | 4. keys

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

The forecast used information from tomorrow

E More workplace conversations / 1-10

Challenge an evaluation shortcut and explain forecast error in the correct units.

An old next-day ticket-volume model used next-day resolved-ticket counts, unavailable when forecasting. A separate rebuilt model uses archived, time-appropriate inputs. On a fixed ten-day holdout unused for training or tuning, its total absolute error is 80 tickets; the agreed baseline totals 100.

PEOPLE / Farah: Forecast analyst | Lewis: Operations planner

Use each bank entry once, without changing its form. Answers follow turn 20.

WORD + PHRASE BANK

data leakage / baseline / mean absolute error / backtest / holdout period / forecast origin

11

- 01 **Lewis:** The old forecast was remarkably accurate. Why have you taken its performance figure out of the planning presentation?
- 02 **Farah:** One input was unavailable at the ¹ _____. It used tomorrow's resolved-ticket count to predict tomorrow's incoming volume.
- 03 **Lewis:** That count was in the historical export, so it looked like an ordinary column. But we would not have known it when making the forecast.
- 04 **Farah:** Exactly. That is ² _____. A field being present in today's historical file does not mean it existed at the earlier decision time.
- 05 **Lewis:** You rebuilt the model without it. Have you also checked when the remaining fields became available, rather than just when their events occurred?
- 06 **Farah:** Yes. This rebuilt evaluation uses archived inputs available at each prediction time, including the actual publication delays. It does not backdate a later data delivery.
- 07 **Lewis:** How did you test it? A random mixture of past and future dates would not look like our daily planning process.
- 08 **Farah:** We used a chronological ³ _____. Training precedes the evaluated predictions, and each forecast uses only the information available at its issue time.
- 09 **Lewis:** The comparison sheet also names a final ten-day block. Was that block used to choose the model settings?
- 10 **Farah:** No. It is the fixed ⁴ _____. The model and comparison rule were settled before those results were examined.

The conversation, continued

E More workplace conversations / 11-20

Continue with the word bank on page 82.

- 11 **Lewis:** Across those ten days the absolute errors add to eighty tickets. What should I call the average, rather than just saying eighty errors?
- 12 **Farah:** The ⁵ _____ is eight tickets per daily forecast. It is in ticket units, not eight percent, and opposite signed errors do not cancel.
- 13 **Lewis:** Our agreed simple forecasting method has total absolute error of one hundred over the same ten days, so its mean is ten tickets.
- 14 **Farah:** Yes. That ⁶ _____ is essential. A model can look sophisticated while failing to improve on a simple alternative.
- 15 **Lewis:** Eight instead of ten is a twenty-percent reduction in mean absolute error for this period. It does not mean eighty-percent forecast accuracy.
- 16 **Farah:** Correct. Also inspect individual days and the operational cost of underestimation versus overestimation. The mean alone does not describe every planning consequence.
- 17 **Lewis:** Can we promise the same improvement next month? Staffing needs a practical expectation, not just the historical result.
- 18 **Farah:** No guarantee. This is a bounded historical evaluation. Changes in demand or input availability can alter performance, so the approved deployment plan needs monitoring.
- 19 **Lewis:** I will present the eight-versus-ten-ticket comparison, name the ten-day period, and explain why the earlier figure was withdrawn.
- 20 **Farah:** Good. Keep the corrected evaluation traceable and the future claim modest. The result supports considering the rebuilt model, not pretending we can use tomorrow's information today.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. forecast origin | 2. data leakage | 3. backtest | 4. holdout period | 5. mean absolute error | 6. baseline

A ticket error is mislabeled as a percentage

F Say it in a new situation

Over eight held-out days, a model has 48 tickets of total absolute error. The reference method has 64 over the same days. No percentage-error metric was calculated.

Complete the four gaps. Use each bank entry once.

WORD + PHRASE BANK

twenty-five / six / eight / error

Analyst: The model mean absolute error is ¹ _____ tickets per forecast.

Planner: The reference method averages ² _____ tickets.

Analyst: That is a ³ _____ percent reduction in this error metric.

Planner: Do not relabel a six-ticket error as six-percent ⁴ _____.

11

ANSWERS / EXPLANATIONS: PAGE 103

1. six | 2. eight | 3. twenty-five | 4. error

Read, switch, repeat

1. Check the four answers and read the complete exchange aloud.
2. Switch roles. Read it again with the word bank covered.
3. Return to the extended conversation. Read turns 1-10, then switch roles for turns 11-20. Use the corrected expressions.

SOLO PRACTICE

Read both roles. Pause briefly at each change of speaker. Repeat any line that needed an answer check.

01 Briefing and language checks

Metric Definitions and Business Questions

A Briefing checks / page 4

1. Which count answers the supplied business question?

A. 900 distinct users with a completed transaction

The question concerns eligible users completing transactions, which the second measure directly counts.

2. What is the correct eligible-user transaction rate?

C. 45%

Nine hundred divided by two thousand eligible users equals forty-five percent.

3. Why can both query results be correct?

D. They apply different event definitions to the same eligible population.

Distinct event definitions can produce different accurate totals without a calculation error.

D Language checks / page 9

1. Which label best fits the 900 count?

A. September transaction-active users

The count concerns distinct users with a completed transaction in the reporting window.

2. What does 900/1,200 measure?

B. The share of logged-in users who completed a transaction

The denominator is the login group, and the numerator is its transacting subset.

3. Why should the two totals not be added?

B. The 900 transacting users are already included among the 1,200 login users.

The supplied subset relationship means adding totals double-counts the transacting group.

4. Which next step addresses the real disagreement?

D. Record both event definitions and select the transaction measure for this question.

The discrepancy is semantic and decision-related, not a demonstrated query failure.

01 Conversation answers

Metric Definitions and Business Questions

E Dialogue gaps

Pages 10-11

- 1. business question.** Turn 2. The business question determines which correctly calculated measure is relevant to the decision.
- 2. qualifying event.** Turn 4. The qualifying event differs between successful login and completed transaction, explaining the distinct totals.
- 3. reporting window.** Turn 6. The reporting window fixes the period and prevents a time-boundary difference from being confused with the event-definition difference.
- 4. distinct user.** Turn 8. A distinct user is counted once even when the user performs multiple qualifying actions within the window.
- 5. denominator.** Turn 10. The denominator defines the rate's base; eligible users and logged-in users answer different percentage questions.
- 6. numerator.** Turn 12. The numerator remains nine hundred transacting users while the stated denominator changes the meaning of the percentage.
- 7. subset.** Turn 14. The subset relationship means every transacting user is already counted within the login group, so addition would duplicate those users.
- 8. semantic layer.** Turn 16. The semantic layer preserves shared business definitions so later reports use the same labels and rules consistently.
- 9. metric owner.** Turn 18. The metric owner is accountable for the definition and its controlled evolution rather than leaving each report to invent its own meaning.
- 10. reconciliation.** Turn 20. Reconciliation explains the different totals and their uses without forcing distinct measures to produce an artificial match.

F Transfer conversation

Page 12

1. 400

Analyst: "The relevant distinct-user count is 400."

The supplied transaction-active group contains four hundred distinct users.

2. 1000

Manager: "The eligible-user denominator is 1000."

The requested rate uses all one thousand eligible users as its base.

3. 40

Analyst: "The requested rate is 40 percent."

Four hundred divided by one thousand equals forty percent.

4. 50

Manager: "Using login users instead would give 50 percent."

Four hundred divided by eight hundred logged-in users equals fifty percent.

02 Briefing and language checks

Data Quality and Trust

A Briefing checks / page 13

1. Which source is complete only through Monday?

A. Refunds

The briefing explicitly limits refund coverage to Monday while orders extend through Tuesday.

2. What is known about Tuesday refunds?

D. They are unknown in the available source.

Missing source coverage does not establish the amount of Tuesday refunds.

3. Which comparison can Asha publish as verified?

C. Monday-only, using aligned source coverage

Both sources are verified complete through Monday, making that shared period usable.

D Language checks / page 18

1. Which dashboard note is accurate?

A. Refreshed Wednesday 09:00 UTC; refunds complete through Monday; two-day net value incomplete.

The note separates successful processing from source coverage and identifies the affected measure.

2. Why is Monday-only reporting usable here?

D. Both sources are verified complete through that shared period.

The relevant fact is aligned verified coverage, not an inherent property of Monday.

3. Which check is insufficient by itself?

A. Seeing that the dashboard refresh job completed

A completed refresh can process stale inputs without repairing their missing coverage.

4. What should happen after the delayed records arrive?

A. Validate and reconcile them before removing the limitation.

Receiving records is followed by verification, and event dates should preserve business timing.

02 Conversation answers

Data Quality and Trust

E Dialogue gaps

Pages 19-20

- 1. refresh timestamp.** Turn 2. The refresh timestamp shows when the dashboard processing completed, not the complete-through date of each input.
- 2. source cutoff.** Turn 4. The source cutoff identifies the period through which the refund source is verified complete, here Monday 23:59 UTC.
- 3. coverage window.** Turn 6. The coverage window differs across orders and refunds, making the combined two-day net measure misaligned.
- 4. zero value.** Turn 8. A zero value asserts a quantity of none; absent Tuesday refund coverage does not support that assertion.
- 5. completeness.** Turn 10. Completeness concerns the required records for the stated period; the briefing verifies both sources for Monday.
- 6. data-quality notice.** Turn 12. The data-quality notice makes the source limitation and affected measure visible where readers interpret the result.
- 7. recovery estimate.** Turn 14. A recovery estimate needs a supported expectation; this case supplies no confirmed time for the missing delivery.
- 8. event time.** Turn 16. Event time records when the business event occurred, which remains distinct from the later ingestion or load time.
- 9. backfill.** Turn 18. The backfill loads previously missing records, which need verification before the incomplete-status notice can be removed.
- 10. timeliness.** Turn 20. Timeliness is availability within the required period; the refund delay should be disclosed while the usable aligned view is preserved.

F Transfer conversation

Page 21

1. Friday

Analyst: "The refresh ran on Friday."

The briefing explicitly identifies Friday as the processing day.

2. Wednesday

Manager: "Returns are complete only through Wednesday."

The source cutoff for returns is Wednesday, not the later refresh day.

3. unknown

Analyst: "Thursday returns are unknown."

Missing Thursday coverage does not prove a zero business amount.

4. aligned

Manager: "A Wednesday-only comparison is aligned."

Both sources are verified complete for Wednesday, so their coverage aligns.

03 Briefing and language checks

Dashboards and Executive Reporting

03

A Briefing checks / page 22

1. What is the September timely-completion rate?

A. 96%

Four hundred eighty timely requests divided by five hundred eligible requests equals ninety-six percent.

2. How does September compare with the target?

B. One percentage point above the 95% target

Ninety-six percent minus ninety-five percent is one percentage point.

3. What was the August rate?

D. 90%

Four hundred fifty timely requests divided by five hundred eligible requests equals ninety percent.

D Language checks / page 27

1. Which card text is most complete?

C. September: 480/500 within 24 hours, 96%; target 95%; August 90%.

The complete text provides the definition, count, denominator, target, and comparable prior rate.

2. Which change statement is accurate?

D. Up 6 percentage points from August

Ninety-six percent minus ninety percent is six percentage points.

3. What is the approximate relative increase from 90% to 96%?

D. 6.67%

The six-point difference divided by the ninety-percent reference rate is approximately 6.67%.

4. What cannot be concluded from the supplied totals?

B. The specific cause of the improvement

The briefing supplies outcomes and comparable definitions but no causal investigation evidence.

03 Conversation answers

Dashboards and Executive Reporting

03

E Dialogue gaps

Pages 28-29

- 1. KPI.** Turn 2. The KPI is the defined performance measure, here the share of eligible requests completed within twenty-four hours.
- 2. actual result.** Turn 4. The actual result is ninety-six percent, calculated from 480 timely completions among 500 eligible requests.
- 3. percentage point.** Turn 6. A percentage point expresses the difference between percentage values, so 96% exceeds 95% by one point.
- 4. comparison period.** Turn 8. The comparison period is the earlier month, whose ninety-percent rate provides the reference for the reported improvement.
- 5. relative change.** Turn 10. Relative change divides the six-point rise by the ninety-percent August rate, giving approximately 6.67%.
- 6. drill-down.** Turn 12. A drill-down exposes the individual or grouped exceptions beneath the favorable summary rate.
- 7. month-on-month.** Turn 14. Month-on-month describes the immediate September-versus-August comparison without claiming a longer established pattern.
- 8. comparability.** Turn 16. Comparability rests on the shared definition, eligibility, and observation basis that make the two monthly rates meaningful to compare.
- 9. observation window.** Turn 18. The observation window must allow the outcome to be observed; the briefing confirms all requests have complete windows.
- 10. executive summary.** Turn 20. The executive summary brings together the result, target, comparison, exceptions, and limits in a concise repeatable account.

F Transfer conversation

Page 30

1. 90

Analyst: "This month is 90 percent."

One hundred eighty divided by two hundred equals ninety percent.

2. 85

Manager: "Last month was 85 percent."

One hundred seventy divided by two hundred equals eighty-five percent.

3. five

Analyst: "The month-on-month increase is five percentage points."

Ninety percent minus eighty-five percent is five percentage points.

4. two

Manager: "This month exceeds target by two percentage points."

Ninety percent minus the eighty-eight-percent target equals two percentage points.

04 Briefing and language checks

Causation, Correlation, and Caveats

04

A Briefing checks / page 31

1. What is the attendee renewal rate?

D. 90%

Ninety renewals divided by one hundred eligible attendees equals ninety percent.

2. What is the non-attendee renewal rate?

C. 80%

One hundred sixty renewals divided by two hundred eligible non-attendees equals eighty percent.

3. Why is the causal effect not established?

B. Attendance was voluntary, and unmeasured baseline differences may matter.

Voluntary selection and possible unmeasured differences prevent the observed comparison from isolating the training effect.

D Language checks / page 36

1. Which headline is supported?

B. Higher observed renewal among voluntary training attendees

The supported headline describes the observed association without asserting an unestablished causal result.

2. What is the relative observed difference compared with 80%?

A. 12.5%

The ten-point gap divided by the eighty-percent reference rate equals 12.5%.

3. Which statement about motivation is accurate?

D. It is an unmeasured possible source of group differences.

The briefing identifies motivation as a possible difference but supplies no measurement or randomization.

4. What should a revised recommendation do?

A. Preserve the association and seek a reviewed design for a causal evaluation.

A better evaluation can address the causal question while the current association remains honestly reported.

04 Conversation answers

Causation, Correlation, and Caveats

E Dialogue gaps

Pages 37-38

1. association. Turn 2. The association is the observed difference between voluntary attendance groups, not an isolated effect of training.

2. self-selection. Turn 4. Self-selection means customers chose attendance, which can create relevant differences between the compared groups.

3. causal effect. Turn 6. The causal effect concerns what training itself changes, which the voluntary comparison does not isolate.

4. Confounding. Turn 8. Confounding is a possible distortion from factors related to both attendance and renewal, not proof that the observed association is meaningless.

5. unmeasured factor. Turn 10. Motivation is an unmeasured factor that could matter, but the case does not establish its size or explanatory contribution.

6. relative lift. Turn 12. Relative lift divides the ten-point observed difference by the eighty-percent comparison rate, giving 12.5% without establishing causation.

7. counterfactual. Turn 14. The counterfactual concerns what the relevant customers would have done under a different training exposure, which is not directly observed here.

8. adjustment. Turn 16. Adjustment can account for specified measured variables but does not automatically remove all unmeasured differences or prove a causal effect.

9. generalizability. Turn 18. Generalizability asks whether a result applies beyond the studied voluntary groups to the broader proposed audience.

10. evaluation design. Turn 20. The evaluation design specifies how the next assessment will address the causal question more credibly than the current voluntary comparison.

F Transfer conversation

Page 39

1. 90

Analyst: "Attendee renewal is 90 percent."

Forty-five divided by fifty equals ninety percent for attendees.

2. 80

Lead: "Non-attendee renewal is 80 percent."

Eighty divided by one hundred equals eighty percent for non-attendees.

3. ten

Analyst: "The observed gap is ten percentage points."

Ninety percent minus eighty percent equals ten percentage points.

4. unestablished

Lead: "The causal effect remains unestablished."

Voluntary attendance and unmeasured baseline differences prevent this comparison from isolating causation.

04

05 Briefing and language checks

SQL, Models, and Transformation Logic

A Briefing checks / page 40

05

1. How many rows result from the supplied join?

D. Three

O1 matches two payment rows and O2 matches one, producing three joined rows.

2. What is the actual total order value?

B. \$200

Two distinct orders worth one hundred dollars each total two hundred dollars.

3. Why does SUM(DISTINCT order_amount) give \$100 here?

D. Both legitimate orders have the same amount, so distinct values collapse them.

Distinct amounts are not distinct orders when different orders can share the same value.

D Language checks / page 45

1. Which operation explains the \$300 result?

D. Summing order amounts after O1 is repeated for two payments

The joined order measure repeats O1's one-hundred-dollar value and inflates the aggregate.

2. Why is summing distinct amounts not a valid repair?

B. It merges different orders that happen to have equal values.

A value-level distinct operation does not deduplicate business entities by their keys.

3. Which proposed repair fits this report?

B. Summarize payments by order ID, then join at one row per order and verify totals.

Pre-aggregation by order ID preserves business identity and aligns the grain. Distinct amounts merge legitimate orders, amount joins use the wrong key, and a left join alone does not prevent multiple matches.

4. Which statement about a left join is accurate?

A. It can preserve unmatched orders but still duplicate an order with multiple matches.

Left retention and match multiplicity are separate properties of the join.

05 Conversation answers

SQL, Models, and Transformation Logic

E Dialogue gaps

Pages 46-47

- 1. row grain.** Turn 2. Row grain describes what each row represents, which changes when an order is matched to multiple payment rows.
- 2. one-to-many relationship.** Turn 4. The one-to-many relationship lets one order match multiple payment records, producing more than one output row for O1.
- 3. fanout.** Turn 6. Fanout multiplies the order row through payment matches, so summing the repeated order measure inflates its total.
- 4. DISTINCT.** Turn 8. DISTINCT on the amount removes equal numeric values, not repeated order identities, so both legitimate hundred-dollar orders collapse.
- 5. aggregation.** Turn 10. Aggregation combines rows at a chosen grain; the measure and that grain must match to avoid repeated order values.
- 6. pre-aggregation.** Turn 12. Pre-aggregation summarizes payments before the join so each order matches at most one summarized payment row.
- 7. join key.** Turn 14. The join key is order ID, which identifies the business relationship rather than a coincidentally equal amount.
- 8. left join.** Turn 16. A left join can retain orders without a matching payment summary, but its retention behavior does not itself prevent fanout.
- 9. reconciliation total.** Turn 18. The reconciliation total is the two-hundred-dollar source order value used to verify that the transformed report preserves the intended measure.
- 10. uniqueness test.** Turn 20. The uniqueness test checks one row per order ID in the summarized data so the corrected relationship can be maintained.

F Transfer conversation

Page 48

1. 160

Engineer: "The correct order total is \$160."

Two distinct eighty-dollar orders total one hundred sixty dollars.

2. 240

Analyst: "The joined sum of order amounts is \$240."

Three joined rows each repeat an eighty-dollar order amount.

3. 80

Engineer: "The sum of distinct amount values is \$80."

The equal eighty-dollar values collapse to one distinct numeric amount.

4. identity

Analyst: "The repair must preserve order identity."

Correct deduplication depends on distinct order identity, not distinct amount values.

05

06 Briefing and language checks

Experiment Readouts and Statistical Thinking

A Briefing checks / page 49

1. What is the point estimate?

C. +1.2 percentage points

The supplied central effect estimate is positive one point two percentage points.

2. Did the lower-bound rule pass?

A. No; -0.4 does not exceed +1.0.

The rule concerns the interval lower bound, not the positive point estimate.

3. What stage has the test reached?

B. Its planned endpoint

The briefing explicitly says the planned endpoint has been reached without new authorization.

D Language checks / page 54

1. Which slide is most faithful to the readout?

D. +1.2 points; 95% interval -0.4 to +2.8; guardrails pass; rollout criterion unmet

The correct slide retains percentage-point units, the negative lower bound, and both conditions of the rollout rule. The alternatives change the decision, scale, or sign.

2. Why does the +1.2 estimate not satisfy the rule?

C. The rule uses the interval lower bound, which is -0.4.

The pre-agreed rule explicitly concerns the lower endpoint exceeding one point.

3. What would be an unsupported next step?

D. Extend the finished test repeatedly until the original method produces a favorable result.

Opportunistic extension changes the analysis behavior without the approved method needed for valid inference.

4. Which statement about zero is correct?

D. Its inclusion does not prove either exact zero effect or equivalence.

An interval containing zero permits uncertainty and does not establish an equivalence conclusion.

06 Conversation answers

Experiment Readouts and Statistical Thinking

E Dialogue gaps

Pages 55-56

1. point estimate. Turn 2. The point estimate is the single plus-1.2-point estimate, which must be interpreted with its uncertainty and the decision rule.

2. confidence interval. Turn 4. The confidence interval communicates uncertainty under its statistical method instead of treating the estimate as a guaranteed effect.

3. lower bound. Turn 6. The lower bound is minus 0.4 points, which fails the requirement that it exceed plus 1.0 point.

4. decision rule. Turn 8. The decision rule requires both the lower-bound condition and passing guardrails, so one passing condition cannot replace the other.

5. no-change value. Turn 10. Zero is the no-change value on the difference scale, but its presence in the interval does not prove the effect is exactly zero.

6. inconclusive result. Turn 12. The result is inconclusive for the desired positive-effect conclusion, while the supplied rollout rule has a definite unmet condition.

7. planned endpoint. Turn 14. The planned endpoint has already been reached, so an opportunistic extension would change the agreed analysis behavior.

8. sequential analysis. Turn 16. Sequential analysis is designed for planned repeated decisions; it cannot be assumed to justify an unapproved extension of this completed test.

9. practical threshold. Turn 18. The practical threshold is the one-point business-relevant boundary used in the agreed lower-bound decision condition.

10. practical significance. Turn 20. Practical significance concerns whether an effect matters for the business, which is not established merely by a positive central estimate.

F Transfer conversation

Page 57

1. 0.8

Analyst: "The point estimate is plus 0.8 points."

The supplied central estimate is positive zero point eight percentage points.

2. 0.2

Director: "The lower bound is minus 0.2 points."

The interval begins at negative zero point two percentage points.

3. 0.5

Analyst: "The required boundary is plus 0.5 points."

The stated lower-bound rule uses positive zero point five as its threshold.

4. unmet

Director: "The rollout condition is unmet."

The negative lower bound does not exceed the required positive threshold.

07 Briefing and language checks

Data Governance and Privacy

A Briefing checks / page 58

1. Which fields directly support the supplied analysis?

B. Week, tier, eligible-case count, and late-case count

The question concerns weekly tier-level rates, which the four specified aggregate fields can support.

2. What is the current approval status?

A. Pending

The briefing explicitly leaves owner approval and review incomplete.

3. What does replacing names with stable IDs establish?

B. It does not automatically establish anonymity.

Stable identifiers may still allow linkage, so name removal alone does not establish anonymity.

D Language checks / page 63

1. Which request best matches the supplied need?

A. Reviewed weekly tier-level counts in the approved workspace, with pending approvals and 30-day access

The correct request supplies the needed measures while preserving the stated handling conditions.

2. Why include eligible-case count?

A. It provides the denominator for each rate.

A late-case count alone cannot show the proportion of eligible cases affected.

3. Which statement about aggregation is accurate?

A. It can reduce detail but still needs the stated disclosure review.

The case explicitly requires review, including small-group risks, despite the aggregate format.

4. What does 30-day access specify?

C. The proposed duration of permission in this case

The briefing defines a local access proposal, not a general retention law or completed approval.

07 Conversation answers

Data Governance and Privacy

E Dialogue gaps

Pages 64-65

- 1. processing purpose.** Turn 2. The processing purpose is the weekly tier-level comparison that determines which fields the analysis needs.
- 2. data minimization.** Turn 4. Data minimization limits the request to sufficient relevant information rather than collecting unrelated personal fields for possible future use.
- 3. aggregation.** Turn 6. Aggregation supplies group-level counts without automatically resolving disclosure risks or approval requirements.
- 4. disclosure review.** Turn 8. Disclosure review assesses sharing risks, including small groups and the proposed audience.
- 5. pseudonymization.** Turn 10. Pseudonymization can replace direct names while preserving possible linkage; it does not automatically establish anonymity or remove controls.
- 6. access scope.** Turn 12. Access scope defines the permitted data and actions, including the approved workspace rather than an assumed unrestricted export.
- 7. Access expiry.** Turn 14. Access expiry ends permission; separate approved rules govern how long the data is retained.
- 8. permitted audience.** Turn 16. The permitted audience identifies who may receive the approved result and prevents internal approval from becoming an unrestricted-sharing assumption.
- 9. secondary use.** Turn 18. Secondary use is a different use beyond the original purpose and may require a new assessment instead of inheriting approval automatically.
- 10. data lineage.** Turn 20. Data lineage records the source and transformations supporting the aggregate counts so their meaning and quality remain traceable.

F Transfer conversation

Page 66

1. count

Analyst: "The rate denominator is the eligible-order count."

Eligible-order count provides the base for the complaint rate.

2. unnecessary

Steward: "Phone numbers are unnecessary for this question."

The briefing explicitly says phone numbers do not support the stated aggregate analysis.

3. 14

Analyst: "The proposed access period is 14 days."

The supplied local proposal specifies fourteen days of access.

4. pending

Steward: "Approval remains pending."

The briefing says access approval has not yet been granted.

08 Briefing and language checks

Insight Storytelling and Recommendations

A Briefing checks / page 67

1. How did the issue-Q ticket rate change?

B. From 2% to 8%, a six-percentage-point rise

Twenty and eighty divided by one thousand give two and eight percent, respectively.

2. What does the mobile count establish?

B. Seventy of the eighty later issue-Q tickets are mobile-related.

The supplied count describes the composition of issue-Q tickets, not a mobile population failure rate.

3. What authorization is requested?

A. A two-day investigation with a subsequent readout

The briefing defines a bounded investigation rather than a fix, cause conclusion, or broader permission.

D Language checks / page 72

1. Which opening best uses the five-minute meeting?

A. Approve a two-day investigation of Q; its ticket rate rose from 2% to 8%.

The opening connects a specific decision to the strongest supplied observation.

2. What is the relative increase from 2% to 8%?

C. 300%

The six-point increase divided by the two-percent original rate equals three hundred percent.

3. Why is 70/80 not a mobile failure rate?

D. Its denominator is issue-Q tickets, not all relevant mobile activity.

The supplied denominator describes issue composition and cannot substitute for the missing mobile base.

4. What does approval of this request authorize?

D. The defined two-day investigation and readout

The stated action boundary limits approval to the investigation, not implementation or causal claims.

08 Conversation answers

Insight Storytelling and Recommendations

E Dialogue gaps

Pages 73-74

1. recommendation. Turn 2. The recommendation is the proposed two-day investigation tied to the documented rise in issue-Q ticket frequency.

2. supporting evidence. Turn 4. The supporting evidence supplies twenty and eighty issue tickets with the same one-thousand-ticket denominator in each week.

3. headline finding. Turn 6. The headline finding states the observed ticket-rate movement with its correct units rather than implying distinct users or a causal explanation.

4. segment concentration. Turn 8. Segment concentration describes where the issue tickets are concentrated, not the failure frequency within all mobile activity.

5. denominator gap. Turn 10. The denominator gap is the missing mobile base required to turn the issue composition into a mobile-specific rate.

6. causal hypothesis. Turn 12. The causal hypothesis is the possible release explanation, which needs evaluation rather than being asserted from timing alone.

7. investigation owner. Turn 14. The investigation owner is the service analyst responsible for the defined inquiry and subsequent evidence readout.

8. time box. Turn 16. The time box limits the investigation duration and creates a review point without guaranteeing a causal conclusion or completed fix.

9. action boundary. Turn 18. The action boundary restricts the approval to investigation and readout, leaving production changes subject to their separate process.

10. decision ask. Turn 20. The decision ask is the precise approval or rejection of the defined two-day inquiry, not endorsement of an unproven cause.

F Transfer conversation

Page 75

1. six

Analyst: "The later issue rate is six percent."

Thirty divided by five hundred equals six percent of all tickets.

2. four

Director: "The increase is four percentage points."

The earlier rate is two percent, so the increase is four points.

3. 80

Analyst: "Mobile represents 80 percent of later issue tickets."

Twenty-four divided by thirty equals eighty percent of issue tickets, not mobile users.

4. investigation

Director: "Approval would authorize an investigation."

The supplied request is for investigation only, not a production modification.

08

09 Conversation answers

Both segments improve, but the overall rate falls

E Extended conversation

Pages 76-77

1. denominator. Turn 2. The denominator is the eligible visit count for the corresponding segment and period; those counts change substantially even though each period totals one thousand visits.

2. percentage points. Turn 4. Percentage points express the difference between the percentage values; both segment rates rise by two points, not by the same relative percentage.

3. mix shift. Turn 8. The mix shift moves a much larger share of visits into lower-converting segment A; changing weights can reverse the direction seen in the separate segment rates.

4. reference weights. Turn 12. The reference weights are the earlier ten-percent share for A and ninety-percent share for B; keeping them fixed creates a separate composition-standardized comparison.

5. standardization. Turn 16. Standardization applies a common weighting basis to improve comparability; it does not prove that an intervention changing traffic would leave every segment's rate unchanged.

6. causal effect. Turn 18. A causal effect concerns what an intervention changes; these aggregate and standardized comparisons alone do not isolate the effect of changing traffic allocation.

F Transfer exchange

Page 78

1. 50

Analyst: The earlier overall rate is 50 percent.

Twenty plus one hundred eighty equals two hundred conversions from four hundred visits, or fifty percent.

2. 40

Director: The later overall rate is 40 percent.

Ninety plus seventy equals one hundred sixty conversions from four hundred visits, or forty percent.

3. ten

Analyst: Each segment improved by ten percentage points.

A rises from twenty to thirty percent and B from sixty to seventy percent, each a ten-point increase.

4. composition

Director: Show the segment sizes so the changing composition is visible.

More visits belong to the lower-converting segment later, so the overall rate can fall despite both within-segment improvements.

10 Conversation answers

The customer moved regions. Did the old sale move too?

E Extended conversation

Pages 79-80

1. current record. Turn 2. The current record contains the customer's latest region, South; using it for every order reclassifies earlier sales instead of preserving their original regional attribution.

2. historical attribution. Turn 4. Historical attribution assigns each fact using the relevant state at its event time; it differs from regrouping all facts under today's customer attributes.

3. Type 2 dimension. Turn 6. A Type 2 dimension preserves successive versions of descriptive attributes so facts can be related to the version valid at the relevant time.

4. surrogate key. Turn 8. A surrogate key identifies the particular warehouse dimension row; the stable customer business identifier can appear on several historical versions.

5. time-based lookup. Turn 12. A time-based lookup matches the fact's relevant date to the dimension version's effective interval, rather than selecting the current version for every fact.

6. restatement. Turn 16. Restatement revises previously reported figures under a documented correction policy; the changed attribution and its reason must remain traceable.

F Transfer exchange

Page 81

1. East

Analyst: The April order belongs to East.

April precedes the May first move, so the order-date region is East.

2. West

Engineer: The June order belongs to West.

June follows the effective move date, making West the relevant historical version.

3. 700

Analyst: The unchanged grand total is 700 dollars.

Three hundred plus four hundred equals seven hundred under either allocation, so the total alone cannot verify the regional split.

4. keys

Engineer: Verify the effective-date intervals and the selected version keys.

Version keys must identify the dimension rows valid on each order date rather than merely the current customer row.

10

Background reading: [Microsoft Learn: Star Schema and Slowly Changing Dimensions](#). The conversation is an original fictional teaching case.

11 Conversation answers

The forecast used information from tomorrow

E Extended conversation

Pages 82-83

1. forecast origin. Turn 2. The forecast origin is the time when the prediction is issued; information generated afterward cannot be treated as an available input to that prediction.

2. data leakage. Turn 4. Data leakage introduces information unavailable for the intended prediction into model development or evaluation, making the reported performance unrepresentative of the real forecasting task.

3. backtest. Turn 8. A backtest evaluates predictions against historical outcomes while recreating the information and timing of the intended forecasting task.

4. holdout period. Turn 10. The holdout period is reserved for evaluation and was not used to fit or tune this model; repeatedly tuning against it would undermine that separation.

5. mean absolute error. Turn 12. Mean absolute error averages the magnitudes of prediction errors without cancellation; eighty tickets of total absolute error over ten days gives eight tickets per daily forecast.

6. baseline. Turn 14. The baseline is the agreed reference forecasting method evaluated on the same days and outcome definition; it gives the new model's error a meaningful comparison.

F Transfer exchange

Page 84

1. six

Analyst: The model mean absolute error is six tickets per forecast.

Forty-eight divided by eight gives six tickets of mean absolute error.

2. eight

Planner: The reference method averages eight tickets.

Sixty-four divided by eight gives eight tickets of mean absolute error.

3. twenty-five

Analyst: That is a twenty-five percent reduction in this error metric.

The reduction is two tickets relative to the eight-ticket reference error: two divided by eight equals twenty-five percent.

4. error

Planner: Do not relabel a six-ticket error as six-percent error.

The metric is in ticket units, and no percentage-error calculation is supplied.

11

Background reading: [scikit-learn: Common Pitfalls and Data Leakage](#). The conversation is an original fictional teaching case.

Find the move you need

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

01 Metric Definitions and Business Questions

Clarify the question: "Are we measuring sign-in or completed transactions?"

Set the window: "Use the September calendar month in UTC."

Choose the relevant count: "Nine hundred users answer the transaction question."

Full phrase set: pages 7-8 | Dialogue: pages 10-11

02 Data Quality and Trust

Acknowledge the refresh: "The dashboard refresh completed at 09:00 UTC."

State the other cutoff: "Orders are verified through Tuesday 23:59 UTC."

Keep the limitation visible: "Place the source-cutoff note beside the figure."

Full phrase set: pages 16-17 | Dialogue: pages 19-20

03 Dashboards and Executive Reporting

Name the measure: "This is the share completed within 24 hours."

State the result: "The timely-completion rate is 96%."

Give the relative alternative: "That is about a 6.67% relative increase."

Full phrase set: pages 25-26 | Dialogue: pages 28-29

04 Causation, Correlation, and Caveats

State the observed rates: "Attendees renewed at 90%; non-attendees at 80%."

Label the finding: "This is an association in the supplied data."

Reject the causal leap: "We cannot attribute the whole difference to training."

Full phrase set: pages 34-35 | Dialogue: pages 37-38

Keep the conversation productive

Use these as precise language models. Replace details only when you know the actual facts, authority, and agreed conditions.

05 SQL, Models, and Transformation Logic

Ask about grain: "What does one row represent before the join?"

Explain the fanout: "The order row is repeated for each matching payment."

Reject the shortcut: "Distinct amounts do not mean distinct orders."

Full phrase set: pages 43-44 | Dialogue: pages 46-47

06 Experiment Readouts and Statistical Thinking

Report the center: "The estimated effect is plus 1.2 percentage points."

Keep the scale explicit: "These are percentage points, not relative percentages."

Separate the conditions: "Passing guardrails does not replace the effect requirement."

Full phrase set: pages 52-53 | Dialogue: pages 55-56

07 Data Governance and Privacy

Define the purpose: "We are comparing weekly late-resolution rates by tier."

Offer an alternative: "Use the reviewed aggregate table."

Bound access: "The proposed access duration is thirty days."

Full phrase set: pages 61-62 | Dialogue: pages 64-65

08 Insight Storytelling and Recommendations

Open with the action: "I recommend a two-day investigation of issue Q."

Give the change: "That is a six-percentage-point increase."

Keep cause open: "The release timing is a lead, not proof of cause."

Full phrase set: pages 70-71 | Dialogue: pages 73-74

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

absolute difference state the absolute difference	33
access expiry enforce access expiry	60
access scope limit access scope	60
accuracy assess data accuracy	14
action boundary state the action boundary	69
active user define active users explicitly	6
actual result report the actual result	23
adjustment assess the adjustment method	33
aggregation control aggregation grain	41
aggregation review aggregation level	59
alternative explanation examine alternative explanations	69
annotation add a relevant annotation	24
anonymization evaluate anonymization	59
approved workspace use the approved workspace	60
association report an association	32
backfill validate a backfill	15
base rate check the base rate	68
baseline characteristic compare baseline characteristics	32
business question clarify the business question	5
calendar month report by calendar month	5
causal claim qualify a causal claim	33
causal effect estimate a causal effect	32
causal hypothesis test the causal hypothesis	68
comparability establish period comparability	24

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

comparison group	select a comparison group	33
comparison period	state the comparison period	23
completeness	verify source completeness	14
confidence interval	interpret a confidence interval	50
confidence level	state the confidence level	50
confounding	assess potential confounding	32
consistency	check data consistency	14
correlation	distinguish correlation from causation	32
COUNT	distinguish COUNT variants	42
counterfactual	define the counterfactual question	33
coverage window	align coverage windows	15
CTE	use a CTE	42
dashboard card	label the dashboard card	23
data catalog	consult the data catalog	60
data governance	establish data governance	59
data lineage	document data lineage	60
data minimization	apply data minimization	59
data model	review the data model	42
data owner	obtain data-owner approval	59
data quality	assess data quality	14
data steward	consult the data steward	59
data-quality notice	publish a data-quality notice	15
decision ask	lead with the decision ask	68
decision log	update the decision log	69

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

decision rule	apply the decision rule	50
deduplication	apply user deduplication	6
denominator	state the denominator	5
denominator gap	disclose the denominator gap	68
descriptive finding	distinguish descriptive findings	68
diagnostic follow-up	scope diagnostic follow-up	69
dimension table	maintain dimension keys	42
direct identifier	remove unnecessary direct identifiers	59
direction of improvement	state the direction of improvement	24
disclosure review	conduct disclosure review	60
DISTINCT	inspect DISTINCT semantics	42
distinct user	count distinct users	5
drill-down	provide a drill-down	24
effect size	assess effect size	50
elapsed time	calculate elapsed time	24
eligible customer	identify eligible customers	33
eligible population	define the eligible population	5
engagement	define customer engagement	32
equivalence	test equivalence appropriately	51
evaluation design	review the evaluation design	33
event count	distinguish event counts from users	5
event time	distinguish event time from load time	14
evidence threshold	define the evidence threshold	69
exception queue	review the exception queue	24

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

exclusion rule	document exclusion rules	6
executive summary	sharpen the executive summary	24
export restriction	enforce export restrictions	60
fact table	define the fact-table grain	42
fanout	detect join fanout	41
filter	apply the filter	42
foreign key	check foreign-key relationships	41
freshness	monitor data freshness	14
generalizability	assess generalizability	33
GROUP BY	group by order ID	41
guardrail	check the guardrails	50
headline finding	sharpen the headline finding	68
identity resolution	review identity resolution	6
inconclusive result	explain an inconclusive result	51
indirect identifier	review indirect identifiers	59
ingestion delay	investigate ingestion delay	15
inner join	inspect inner-join matches	42
insight	state the insight	68
internal account	exclude internal accounts	6
interval estimate	report an interval estimate	50
investigation owner	assign the investigation owner	69
issue category	validate the issue category	68
join cardinality	inspect join cardinality	41
join key	choose the join key	41

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

KPI define a KPI	23
late-arriving data handle late-arriving data	15
leading question avoid leading questions	69
left join preserve orders with a left join	42
load time record load time	14
login-based activity report login-based activity	6
lower bound compare the lower bound	50
metric define a metric	5
metric definition agree a metric definition	5
metric owner identify the metric owner	6
minimum detectable effect specify the minimum detectable effect	51
month-on-month describe month-on-month movement	23
motivation assess baseline motivation	32
no-change value identify the no-change value	50
null handling define null handling	42
null hypothesis state the null hypothesis	51
null value distinguish null from zero	15
numerator verify the numerator	5
observation window complete the observation window	24
observational study interpret an observational study	32
one-to-many relationship model a one-to-many relationship	41
percentage point report percentage-point changes	23
performance target compare with the performance target	23
permitted audience define the permitted audience	60

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

personal data	assess personal-data handling	59
pipeline	monitor the data pipeline	15
planned endpoint	respect the planned endpoint	51
point estimate	report the point estimate	50
practical significance	assess practical significance	51
practical threshold	set a practical threshold	50
pre-aggregation	apply pre-aggregation	41
primary key	validate the primary key	41
primary outcome	specify the primary outcome	51
priority criterion	apply the priority criterion	69
processing purpose	state the processing purpose	60
production change	approve a production change	69
pseudonymization	assess pseudonymization safeguards	59
purpose limitation	document the analysis purpose	59
qualifying event	specify the qualifying event	5
random assignment	use random assignment	32
randomized experiment	interpret a randomized experiment	50
re-identification risk	assess re-identification risk	59
readout	schedule the readout	69
recommendation	frame the recommendation	68
reconciliation	reconcile metric totals	6
reconciliation check	run reconciliation checks	15
reconciliation total	compare reconciliation totals	42
recovery estimate	qualify the recovery estimate	15

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

refresh timestamp display the refresh timestamp	14
relative change calculate relative change	23
relative lift qualify relative lift	33
release timing compare release timing	69
renewal measure customer renewal	33
renewal window align renewal windows	33
reporting window align reporting windows	5
retention period document the retention period	60
row grain define the row grain	41
run chart review the run chart	24
sampling variability account for sampling variability	51
seasonality assess seasonality	24
secondary use review proposed secondary use	60
segment concentration report segment concentration	68
selection bias evaluate selection bias	32
self-selection account for self-selection	32
semantic layer maintain a semantic layer	6
sequential analysis use a valid sequential analysis	51
service-level target track the service-level target	23
small cell review small cells	60
source cutoff state the source cutoff	14
SQL review SQL logic	41
standard error report the standard error	51
statistical power plan statistical power	51

Find a term and its context

The page number leads to the definition and a typical word combination. Repeated terms have separate entries when they serve different lesson contexts.

statistical significance	qualify statistical significance	51
subset	identify the subset relationship	6
SUM	check the SUM result	41
supporting evidence	present supporting evidence	68
test account	filter test accounts	6
threshold	apply the threshold	24
ticket mix	analyze the ticket mix	68
time box	set a time box	69
time zone	fix the reporting time zone	5
timeliness	define timeliness requirements	14
timely completion	measure timely completion	23
transaction-based activity	report transaction-based activity	6
treatment group	define the treatment group	32
trend	qualify a trend claim	24
uniqueness	test record uniqueness	14
uniqueness test	run a uniqueness test	42
unmeasured factor	disclose unmeasured factors	33
upper bound	report the upper bound	50
upstream source	inspect upstream sources	15
validity	test data validity	14
variance	explain the target variance	23
watermark	inspect the pipeline watermark	15
year-on-year	report year-on-year change	23
zero value	preserve zero values correctly	15

Original cases. Clear boundaries.

These are original fictional teaching conversations, not transcripts of real workers. Names, figures, organizations, and incidents are invented. Sources support terminology and communication aims, not the fictional results or an endorsement of this book.

[1] UK Government. The Government Data Quality Framework.

[Open the original source](#)

Background for fit-for-purpose data, quality dimensions, and communicating limitations. The dashboard sources and freshness criteria here are fictional. Checked 30 September 2026.

[2] PostgreSQL Documentation. Table Expressions.

[Open the original source](#)

Background for join and grouping behavior. The order-and-payment examples are original, small teaching datasets rather than production queries. Checked 30 September 2026.

[3] NIST/SEMATECH e-Handbook. What Are Confidence Intervals?

[Open the original source](#)

Background for interval estimates and uncertainty language. The supplied experiment interval is fictional and is interpreted, not estimated from raw observations. Checked 30 September 2026.

[4] Information Commissioner's Office. Principle (c): Data Minimisation.

[Open the original source](#)

Background for purpose-linked data selection. This UK guidance is under review; the exercises use explicit fictional local rules, not universal legal conclusions. Checked 30 September 2026.

Phil Stilwell

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